

CPT202 Assignment 2

Group Report for Software Engineering Group Project

2024/2025 Semester 2
An online meeting booking system
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Group number: 17

Submitted by: < Xiangyi Zhu, 2251651 >

Group Members:

1. < Xiangyi Zhu, 2251651 >
2. < Xinrong Shi, 2251997 >
3. < Zimo Zheng, 2147369 >
4. < Nan Lin, 2252173 >
5. < Xiaoya Li, 2255207 >
6. < Ziwei Zhou, 2253073 >
7. < Jiayi Wang, 2255986 >
8. < Hanqiu Zhang, 2251903 >
9. < Xinyuan Zhang, 2255403 >

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1. Introduction

1.1. Background and Requirements

In recent years, the increasing complexity of academic and administrative activities within schools has led to a significant increase in the demand for online meeting room booking systems. This trend highlights the urgent need for efficient resource management and convenient collaboration tools. However, many schools' existing meeting room booking systems still face many challenges, including complex user interfaces, inefficient booking processes, insufficient conflict management, and lack of real-time notification functions. These issues often lead to poor user experience, low resource utilization, and inefficient communication (Sanchez *et al.*, 2019).

In order to effectively address these challenges, it is essential to develop a robust and user-friendly school meeting room booking system. The system should be designed with users in mind, provide an intuitive interface and simplify the booking process, support real-time resource status updates to prevent conflicts, and integrate automatic notification functions to ensure that users receive booking confirmations and reminders in a timely manner. In addition, the system should also be seamlessly connected with existing school scheduling, payment systems, and permission management tools to comprehensively improve administrative efficiency (Shili *et al.*, 2009).

By optimizing resource allocation and improving user experience, the system can not only help schools manage meeting room resources more efficiently, but also provide teachers and students with a smoother collaborative environment, thereby promoting the orderly development of academic and administrative activities.

1.2. Aims of The Project

The primary objective of this project is to design and implement an intuitive web-based school meeting room booking system:

- **For users** (e.g., students and teachers), they can register for an account to become platform users. The system provides a simple and user-friendly interface for viewing the availability of meeting rooms, submitting booking requests, and receiving real-time notifications and confirmations.
- **For administrators**, the system offers a management backend for monitoring and managing meeting room resources. This backend supports a range of functions, including real-time updates on meeting room status, automatic detection of booking conflicts, user permission management, and data visualization analysis. These features help administrators allocate resources more efficiently, optimize meeting room usage, and support the school's teaching and administrative activities.

1.3. User Characteristics

The system supports three main user roles: ordinary users, administrators, and super administrators.

Ordinary users can log in to the system to view meeting rooms, submit reservation applications, view reservation history, cancel reservations, submit maintenance requests, and view notifications. Administrators are responsible for approving reservation applications, monitoring meeting room usage, handling maintenance requests, managing meeting room status, and issuing internal notifications.

As the highest-level operator of the system, the super administrator is responsible for maintaining data integrity and system performance. His main permissions include managing user roles, configuring meeting room classifications, adding, updating or deactivating meeting rooms, maintaining meeting room information, managing permission allocation, and reviewing system logs.

The three roles work together to ensure the efficient operation of the system and achieve the orderly advancement of resource optimization and administrative supervision.

1.4. Project Scope

The scope of the project is illustrated in the Figure 1:

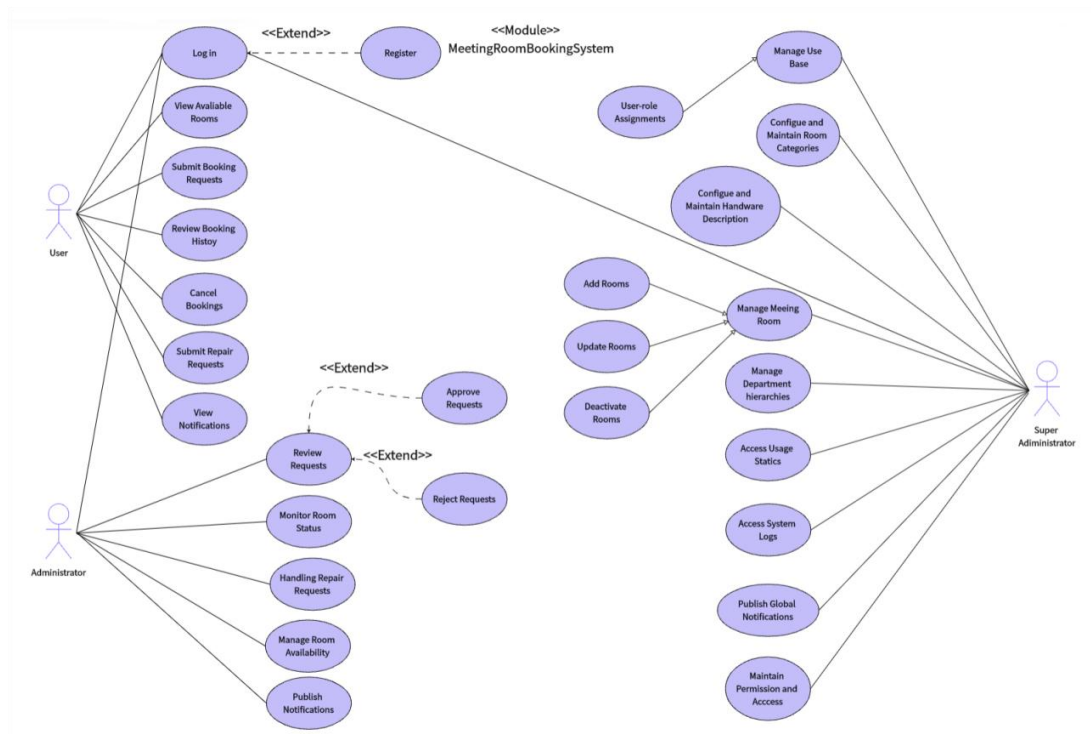


Figure 1 / APPENDIX 1 : Use Case Diagram of the System

1.5. Assumptions and Dependencies

The successful implementation of the project depends on a series of prerequisites and external environmental factors, such as resource availability, technical support, team collaboration, and data integrity. In addition, the project also depends on the stability of third-party API services (e.g., email notifications), reliable network connections, and the fulfillment of external service agreements. During the project implementation process, these prerequisites and dependencies need to be continuously monitored and managed to reduce potential risks and ensure the smooth achievement of project goals.

1.6. Project Risks

Although this project has brought many opportunities in terms of meeting room booking management and improving team reporting and presentation efficiency, there are also some risks that need to be taken seriously during the implementation process. These risks include uneven resource allocation, technical challenges such as system integration and data processing, potential security risks, and rigorous testing that must be carried out to ensure system stability and performance. In addition, the smooth implementation of the project also depends on good collaboration between teams, sufficient supply of resources, and stable support from external services. Throughout the project process, these factors need to be continuously paid attention to and managed to effectively reduce risks and ensure the smooth realization of project goals.

2. Architectural Design

The frontend needs to be responsible for page rendering and interactive experience optimization, while the backend needs to independently provide standardized API interfaces to support flexible calls. Therefore, in order to support multi-terminal access, achieve flexible expansion, improve development efficiency and maintain scalability, the system architecture must adopt a frontend and backend separation mode. Our meeting room

booking system adopts a hierarchical model-view-controller (MVC) architecture design to achieve clear separation of concerns, modular scalability and easy maintenance.

2.1. System Architecture

2.1.1 Implementation in Our Project

- **Model Layer:**

Due to the need to implement important functions such as role permission mapping control and meeting room availability check, our model layer uses MySQL and MyBatisPlus and MyBatisPlusJoin as ORM frameworks. Druid provides efficient database connection pools and monitoring, simplifies multi-table links and improves efficiency. The table follows the third normal form (3NF) with foreign keys to ensure the integrity between key entities.

- **View Layer:**

Our frontend is built with Vue.js 3 (Shavin, 2023) and TypeScript to provide type safety, including components such as FullCalendar for displaying meeting room availability and ECharts for visualizing usage analytics. Element Plus provides consistent and responsive UI components for different modules such as booking approval and meeting room filtering. Vue Router manages access to different views based on user roles.

- **Controller Layer:**

Spring Boot facilitates RESTful API development, handling HTTP requests, and orchestrating business logic across services and mappers (Heckler, 2021). APIs are logically grouped by function, such as meeting room management, user role management, and booking approval workflow.

Since role and permission management requires the role-based access control (RBAC) system, we need to use Sa-Token and Spring Security to help us improve the design of the permission verification layer. Among them, Sa-Token is responsible for managing authentication and role-based access, and password encryption is protected by Spring Security Crypto.

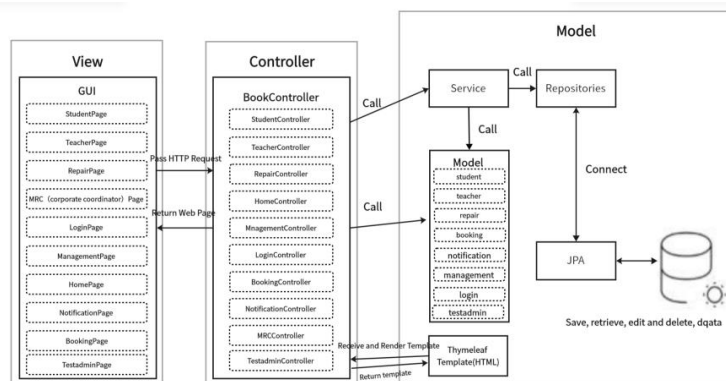


Figure 2 / APPENDIX 2 : MVC architecture

2.1.2 Justification for Choosing Technology

The model layer does not use JPA/Hibernate because its performance in multi-table dynamic query is weak and it is difficult to meet business needs. In contrast, MyBatis-Plus is more flexible and efficient. MySQL operation and maintenance are simpler than PostgreSQL. Druid supports visual monitoring and is easier to debug and optimize than HikariCP.

In the view layer, we choose Vue.js and Element Plus because their syntax is more intuitive, development efficiency is higher, and the interface style is simple, which is adapted to the management system requirements; in comparison, React has a higher cost to get started, Ant Design Vue style is heavy, and Chart.js is difficult to meet complex interactions.

The control layer does not use Spring Security alone because its configuration is complex and difficult to get started. Sa-Token is lighter and suitable for rapid integration of role permission control, so we combine these two techniques. The Java thread model is better than Node.js and is more suitable for processing computationally intensive scenarios such as meeting room conflict detection. Swagger is not introduced in the interface document for the time being, because the current built-in OpenAPI has met the requirements.

2.2. Frontend System

Since the system frontend needs to implement dynamic data display, complex form interaction and real-time status update (such as meeting room reservation, booking approval, chart display, etc.), it has high requirements for response speed and user experience, so we chose Vue.js, which is lightweight, efficient, supports componentization and two-way binding. Compared with React and Angular, its learning cost and configuration complexity are lower, and it is more suitable for short-term delivery projects, allowing developers to focus on business logic rather than framework details.

Core Technologies and Frameworks:

Based on the Vue framework we chose, we give priority to using tools that are highly integrated with it to reduce configuration costs and improve development efficiency. Vite is deeply integrated with Vue, and there is no need to manually configure Loader, which is better than the complex configuration of Webpack. Vue Router natively supports role-based permission routing, with concise syntax, easy to develop quickly, and reduces learning costs. Axios can unify the error and permission logic, and simplify the request cancellation mechanism, avoiding the need to manually encapsulate AbortController when using the Fetch API.

UI/UX & Optimization:

In terms of UI component selection, Element Plus has a simple style, which meets the needs of the meeting booking system and is better than Ant Design Vue, which has a heavier style. FullCalendar has intuitive configuration, supports theme customization, and is more suitable for time interaction than Tui Calendar. ECharts provides rich charts and powerful interactions, which is better than the limited Chart.js and meets the needs of multi-dimensional data visualization. To improve the user experience, we introduced Vue Tippy to provide adaptive contextual prompts, replacing the native Title to achieve more flexible teaching guidance. At the same time, the system implements role-based routing control through Vue, and supports conditional filtering and paging to display meeting room information.

It is worth mentioning that for MRC and Test-Admin users, Vue's <keep-alive> keeps the management panel persistently activated, which is lighter than manual state management (such as Vuex) and improves the continuous operation experience.

2.3. Backend System

In order to support multi-terminal access and improve the interactive experience, this system adopts a frontend and backend separation architecture: the frontend is responsible for page rendering and user interaction, and the backend provides a standardized RESTful API to support flexible calls. Unlike traditional JSP template rendering, the decoupling of the frontend and backend can significantly improve development efficiency, module scalability, and deployment flexibility, and is particularly suitable for fast-iteration project requirements. The backend architecture is based on Spring Boot's MVC layered model to achieve the division of responsibilities for control logic, business logic, and data access, enhancing the maintainability and testability of the system. Compared with lightweight solutions such as Express (Node.js), Java's strong typing characteristics and thread model are more suitable for business-intensive scenarios such as booking conflict detection and approval flow control in this system.

Controller Layer:

The Controller layer, built with Spring Boot's @RestController, maps incoming HTTP requests to corresponding service operations, exposing APIs for critical functionalities such as meeting room booking management, room availability maintenance, user and department administration, repair issue tracking, and room usage statistics visualization.

Spring Security enforces request-level authentication and protects sensitive endpoints (e.g., booking approvals, user role modifications), while Sa-Token manages token-based authentication, session persistence, and fine-grained role authorization across different user types, ensuring secure and context-aware access for Students, Teachers, MRC Coordinators, Repair Staff, and Super Admins.

Service Layer:

The Service layer encapsulates the business logic of the Meeting Booking System, acting as a bridge between controllers and data persistence layers. It validates meeting room booking requests, manages approval workflows for MRC users, processes room status updates (e.g., availability, maintenance), handles user authentication and

secure password management (via Sa-Token and Spring Security Crypto), generates room usage statistics for administrative dashboards, and tracks repair submissions and resolutions.

The use of @Service annotations and dependency injection supports modular business design, improves maintainability, and facilitates unit testing across key service modules.

DAO (Data Access Object) Layer:

The DAO layer abstracts database interactions using MyBatis-Plus and MyBatis-Plus-Join, simplifying CRUD operations and multi-table joins. This approach reduces boilerplate SQL code, accelerates development, and minimizes potential database inconsistencies. Key operations include managing meeting rooms, booking records, user-role mappings, and statistical queries.

By encapsulating database logic, this layer provides a reusable and centralized structure for persistence access (Li *et al.*, 2020).

Entity Layer:

The Entity layer defines Java classes mapped directly to relational tables within a MySQL database. Entities such as sys_user, meeting_room, meeting_book, sys_dept, and repair represent core business objects. Annotations like @TableName and @TableField ensure tight coupling between object models and database schema, while foreign keys and association tables maintain data integrity.

This well-defined backend architecture supports efficient request handling, secure data management, and the modular scalability required for the ongoing evolution of the Meeting Booking System.

2.4. High-level Database Design

Our database design utilizes MySQL to support the persistent storage and logical organization of core entities in the Meeting Booking System. The schema, normalized through an Entity-Relationship Diagram (ERD), ensures a scalable and consistent data model aligned with the system's operational logic. Table relationships are maintained through foreign key constraints to enforce referential integrity (DuBois, 2013). This design ensures minimal redundancy, optimized query performance, and high data integrity.

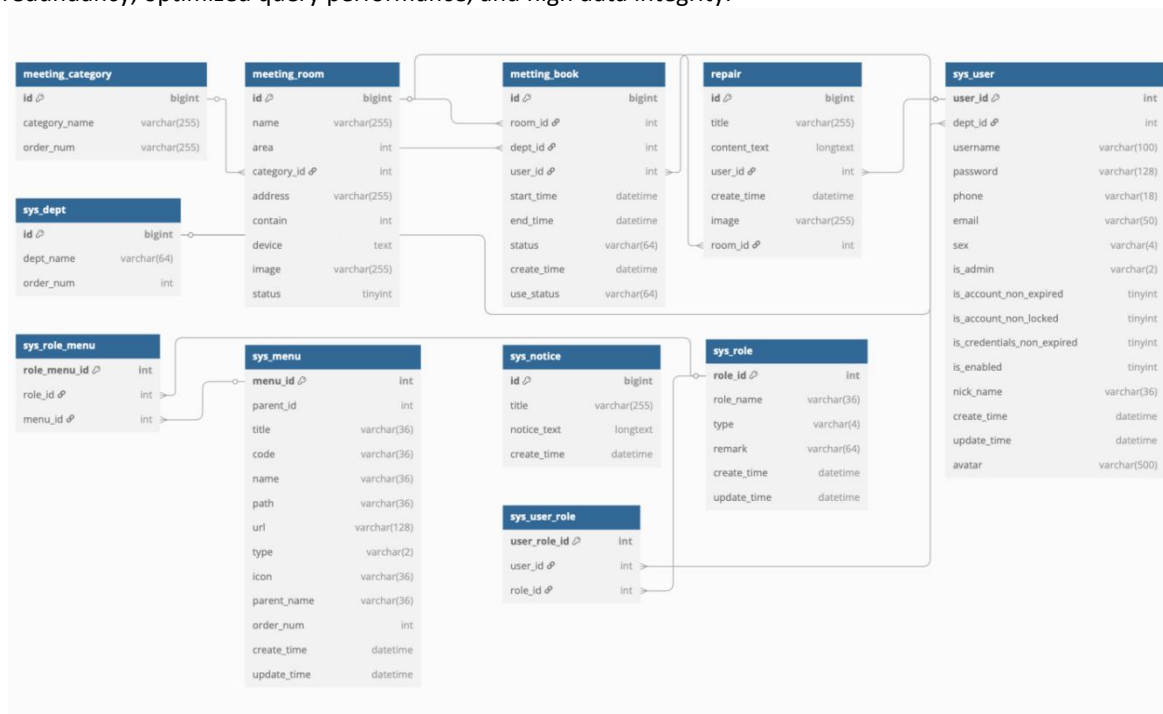


Figure 3 / APPENDIX 3 : ER Diagram

Entity Identification:

These entities form the framework of the system, supporting multi-role interactions, scheduling processes, access management, and communication flows.

Data Normalization:

We followed the third normal form (3NF) principle in database design to ensure data independence, reduce redundancy, and maintain data relevance. By dividing data of different topics into independent tables and setting foreign key relationships, the system can maintain consistency when updating data, and daily maintenance is more efficient. The final entity relationship diagram (ERD) matches the modular design of the overall system, ensuring that each data table can directly correspond to the relevant service interface, while laying the foundation for subsequent function expansion and system upgrades (Mendoza and Piedra, 2020).

3. Software Design

3.1. Software Modules

In the early stage of design, we considered the actual situation and set up five core modules for the meeting room reservation system divided by role, covering students, teachers, MRC, maintenance personnel and super administrators. If this solution cannot meet the expansion needs, our alternative solution is to divide the modules by business function to achieve higher independence and scalability between modules, which is convenient for the introduction of new functions or plug-ins later. However, this solution is complex in design and is more suitable for the long-term evolution of medium and large projects.

1. Student and Teacher Module: Students and teachers serve as standard users, able to browse available meeting rooms, submit booking requests, view booking histories, report maintenance issues, and access announcements and bulletins. Teachers may have additional privileges depending on institutional policies, but both roles primarily focus on personal meeting management.

2. MRC (Conference Coordinator) Module: MRC users oversee meeting room scheduling, reviewing and approving booking requests, managing room availability and categories, tracking repair records, and monitoring room usage patterns. This module ensures the orderly allocation and efficient utilization of meeting resources.

3. Repair Staff Module: Repair staff are responsible for processing maintenance requests. They can view submitted repair reports, update repair statuses (e.g., pending, in progress, completed), and manage supporting documentation such as images or maintenance notes through a task-focused dashboard.

4. Super Admin Module: Super Admins possess full administrative privileges, including user, department, and role management, system configuration, menu permission settings, and announcement publishing. They are responsible for system-wide oversight, deployment, and high-level maintenance.

In addition to role-specific modules, the system provides reusable core functionalities such as the Announcement and Repair Management modules, which offer consistent user experiences across roles while maintaining clear functional segregation (Farouk, 2017).

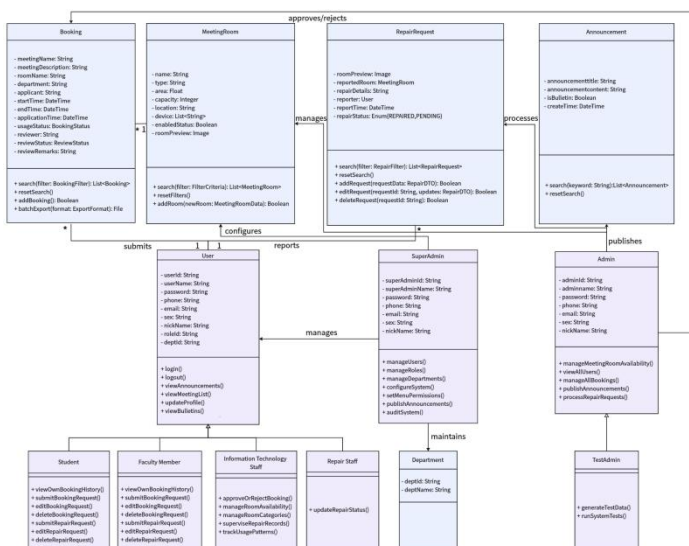


Figure 4 / APPENDIX 4 : Software Modules Design

Example: Role-Based Menu Permission Code Snippet:

```
List<SysMenu> menuList = "1".equals(user.getIsAdmin())
    ? sysMenuService.list()
    : sysMenuService.getMenuByUserId(user.getUserId());

List<String> collect = menuList.stream()
    .filter(item -> item != null &&
        StringUtils.isEmpty(item.getCode()))
    .map(sysMenu->getCode())
    .collect(Collectors.toList());
```

Figure 5 : Brief Code of Role-Based Menu Permission

```
// Initialize an empty menu list
List<SysMenu> menuList = null;

// Check if the user is a super administrator
if (StringUtils.isEmpty(user.getIsAdmin()) && "1".equals(user.getIsAdmin())) {
    // Super administrator, directly query all menus
    menuList = sysMenuService.list();
} else {
    // For regular users, get menus assigned to their user ID
    menuList = sysMenuService.getMenuByUserId(user.getUserId());
}

// Validate if the user has any valid menu permissions
if (menuList.stream().filter(Objects::nonNull).collect(Collectors.toList()).isEmpty()) {
    return ResultUtils.error("The user's role has not been assigned menu permissions, Please log in with an administrator account to assign menus.");
}

// Set the 'code' field from the menu table
List<String> collect = Optional.ofNullable(menuList).orElse(new ArrayList<>()) // Handle potential null list, use empty list if null
    .stream()
    .filter(item -> item != null && StringUtils.isEmpty(item.getCode())) // Filter items with empty codes
    .map(sysMenu->getCode()) // Extract code field
    .collect(Collectors.toList()); // Collect to string list
```

Figure 6 / APPENDIX 5 : Role-Based Menu Permission Code

From a modular design perspective, each module exhibits high cohesion — with related business logic tightly grouped - and minimal interdependencies, ensuring scalability, maintainability, and ease of future expansion.

3.2. High-level Process Flow

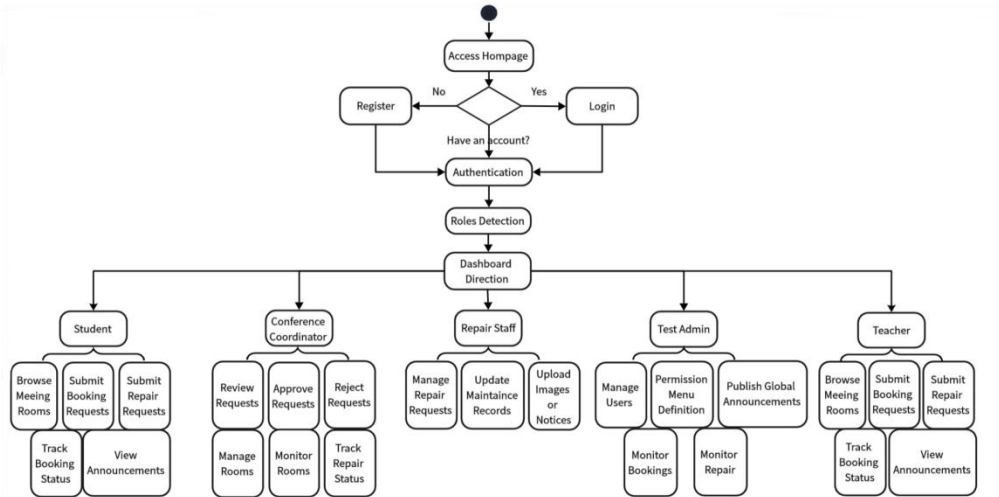


Figure 7 / APPENDIX 6 : Process Flow

The Meeting Booking System integrates a full suite of core functionalities for managing institutional meeting room reservations, with each module logically separated and collaboratively coordinated to streamline both user experience and administrative oversight.

Entry and Authentication Flow :

Upon accessing the system homepage, users can register by submitting account credentials and basic personal information. Following successful registration and login, the system enforces secure session management, including duplicate-login prevention and session expiration. Based on authenticated roles, users are automatically redirected to their respective dashboards.

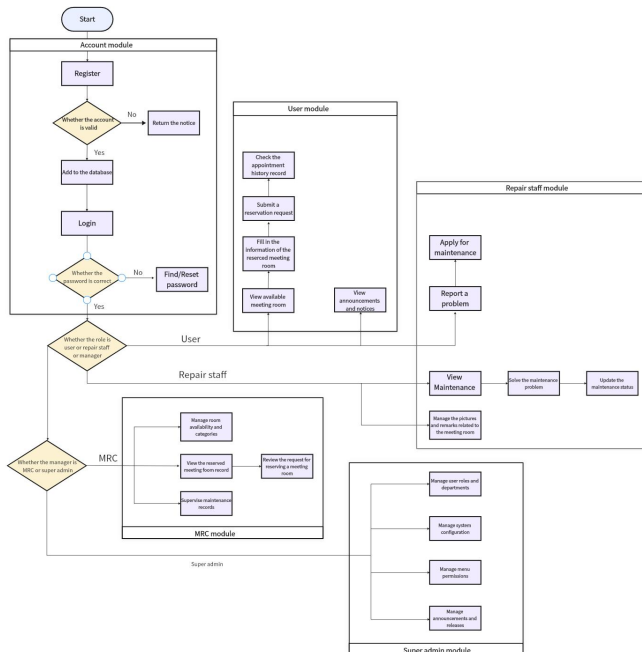


Figure 8 / APPENDIX 7 : Role login division flow chart

```
//Login
@PostMapping("/login")
public ResultVo login(HttpServletRequest request, @RequestBody LoginParam param) {
    //Get the data passed from the frontend
    String code = param.getCode();
    //Acquire session
    HttpSession session = request.getSession();
    System.out.println("Session ID: " + session.getId());
    System.out.println("Session Creation Time: " + new Date(session.getCreationTime()));
    System.out.println("Session Last Access Time: " + new Date(session.getLastAccessedTime()));

    //Acquire session id code
    String code1 = (String) session.getAttribute("code");
    System.out.println("Code in session: " + code1);

    // If (StringUtil.isEmpty(code1)) {
    //     return ResultUtils.error("Verification code expired.");
    // }
    // Determine the incoming frontend parameters code and session is the content inside equal?
    // If (!code1.equals(code)) {
    //     return ResultUtils.error("Verification code is incorrect.");
    // }
}
```

Figure 9 / APPENDIX 8 : Login Authentication flow chart code (1)

```
QueryWrapper<SysUser> query = new QueryWrapper<>();
query.lambda().eq(SysUser::getUsername, param.getUsername());
SysUser user = sysUserService.getOne(query);
if (null == user) {
    return ResultUtils.error("Username error!");
}
if (!passwordEncoder.matches(param.getPassword(), user.getPassword())) {
    return ResultUtils.error("Wrong password!");
}
if (!user.isEnabled()) {
    return ResultUtils.error("Account has been banned, please contact the system administrator!");
}
StpUtil.login(user.getUserId(), "PC");
//Return user information and token
LoginVo vo = new LoginVo();
vo.setUserId(user.getUserId());
vo.setUserName(user.getUserName());
//Generate token
// FirstStep: Acquire Token Related parameters
SaTokenInfo tokenInfo = StpUtil.getTokenInfo();
vo.setToken(tokenInfo.getTokenValue());
return ResultUtils.success("Login successful", vo);
}
```

Figure 10 / APPENDIX 8 : Login Authentication flow chart code (2)

Role-Based Routing and Functional Flow

- **Student / Teacher Users** are directed to a personalized dashboard, where they can browse available meeting rooms via a calendar interface (integrated with FullCalendar), submit and manage booking requests, track booking history, report facility issues, and view system announcements.
- **MRC (Conference Coordinator)** access an administrative dashboard enabling them to review, approve, or reject booking applications, manage room availability and metadata, monitor room usage trends, oversee repair ticket statuses, and interact with a persistent information pane implemented via Vue 3, Element Plus, and <keep-alive>, ensuring critical approval tasks remain continuously visible.
- **Repair Staffs** are routed to the repair management module, where they can view user-submitted maintenance reports, update repair statuses (e.g., pending, in progress, completed), upload supporting documentation, and prioritize repair workflows based on task urgency.
- **Super Admin** users gain full administrative access, managing system-wide entities including users, departments, roles, and permission menus (sys_menu), publishing global announcements (sys_notice), monitoring all booking and repair activities, and operating within a persistent admin workspace enhanced by router caching and <keep-alive> to maintain navigation state and operational continuity.

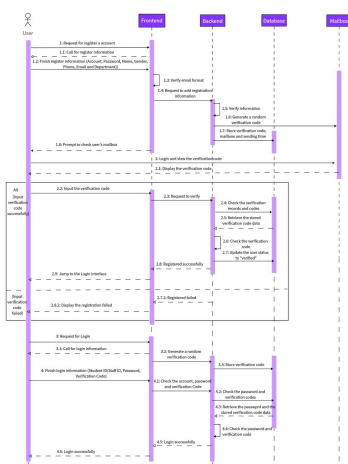


Figure 11 / APPENDIX 9 : User registration and login

This role-driven, modular process flow supports high usability, system reliability, and functional scalability across diverse user scenarios.

3.3. Software Support Services

Our Meeting Booking System relies on a suite of software tools and service integrations that collectively enhance development efficiency, system performance, and operational maintainability. These services span across database development, interface navigation, and other supporting services.

Database-Related Services

• MySQL:

MySQL serves as the primary relational database management system, supporting normalized table structures and high-performance query execution. Foreign key constraints ensure robust referential integrity between core entities such as users, meeting rooms, reservations, and permissions.

• MyBatis-Plus + MyBatis-Plus-Join:

We adopted MyBatis-Plus and MyBatis-Plus-Join as our ORM solution to streamline SQL operations. These tools allow concise query writing, auto-generation of boilerplate code, and advanced conditional query support, significantly reducing the complexity and redundancy of database access logic.

• Navicat:

Navicat was used during the development phase to visually manage the MySQL database. It enabled collaborative access to table structures, ERD diagrams, test queries, and data population across development environments (Ozar, 2012).

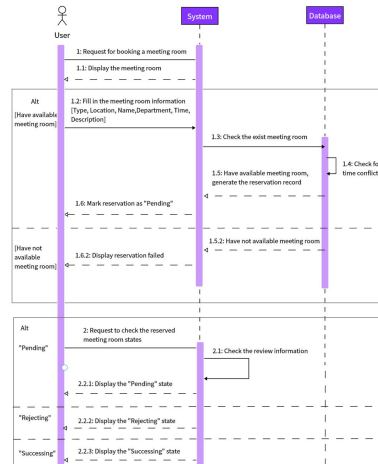


Figure 12 / APPENDIX 9 : User reserved meeting room

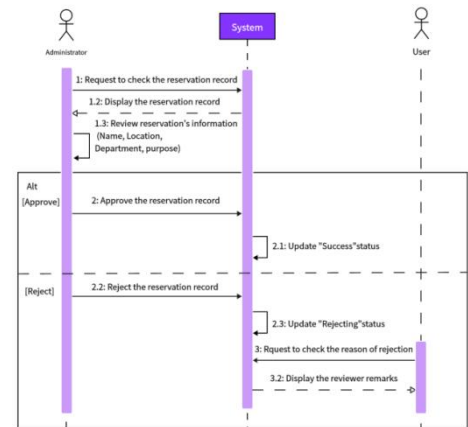


Figure 13 / APPENDIX 9 : Administrator reviews meeting room booking records

Web Navigation Services

The system features a role-adaptive sidebar navigation built with Vue.js 3 and Element Plus, supporting collapsible, hierarchical menus with route-based access control. Navigation state is persistently maintained using Vue Router and <keep-alive> to ensure smooth user experiences across modules. This responsive layout guarantees accessibility across varied screen resolutions and supports interface consistency for students, teachers, MRCs, repair staff, and super administrators.

Other Supporting Services

- **Spring Boot Mail Service:**

The system integrates Spring Boot's JavaMailSender for email-based verification. During registration, the backend generates a secure, randomized code and sends it to the user's provided email. The mail service is designed to be event-driven and extensible for future use cases such as reservation status notifications and administrative announcements (Deinum, 2024).

- **Git + Coding.net:**

Source code versioning was managed via Git, with remote repositories hosted on Coding.net. Each team member developed within independent branches following a Gitflow-like model, using pull requests for integration and code reviews. Coding.net also supported Scrum task tracking and Wiki documentation (Guthals, 2023).

- **Maven:**

Maven was used as the backend's dependency and build management tool. It helped manage transitive dependencies, resolve version conflicts, and package the Spring Boot application into a JAR file for production deployment. Maven's standard project structure also facilitated better collaboration among backend developers (Varanasi, 2019).

Together, these services form a cohesive foundation that supports rapid development, modular deployment, and stable maintenance throughout the project lifecycle.

3.4. Coding Structure and Convention

Project Structure

The backend project is organized following a conventional multi-layered architecture, with packages segmented into controller, service, mapper, entity, config, and utils, aligned with Model-Service-Controller design principles. Detailed definitions of each directory's responsibilities are documented internally (**refer to APPENDIX 10, for the Java directory structure**).

The frontend project, built with Vue.js 3 and Vite, adopts a modular structure, separating source code into folders such as views, components, router, assets, store, and api, supporting component-based development and role-based route control.

Method Invocation Convention

Controllers invoke service methods strictly through defined interfaces, ensuring loose coupling between layers. Each interface may have one or more implementation classes, providing flexibility for logic extension and supporting unit test mockability.

Naming Convention

We adhered to standardized naming practices for consistency and clarity:

1. **Methods and variables:** camelCase (e.g., createBookingRequest)
2. **Class names:** PascalCase (e.g., BookingController)
3. **Constants:** UPPER_CASE with underscores (e.g., MAX_ROOM_CAPACITY)
4. **Naming principle:** all names are semantically meaningful and follow the "entity + action" pattern when applicable (e.g., getUserPermissionList)
5. **Interface implementation classes:** suffixed with Impl (e.g., RoomServiceImpl)

Code Maintenance Standards

During collaborative development, code maintainability was emphasized:

1. Developers are expected to write self-documenting code with concise and relevant inline comments
2. Prior to any Git commits, unused code segments and outdated comments are cleaned up to preserve repository hygiene
3. Code formatting follows IDE-configured style guides across the team to maintain structural consistency

Task Management and Code Completion

Unfinished features and pending development tasks are systematically tracked using Coding.net's Scrum board and Wiki documentation, rather than through inline TODO comments. Sprint-based tracking ensures transparent task assignment and facilitates continuous development progress.

By adhering to these standardized structures and conventions, the Meeting Booking System achieves high code readability, efficient collaboration, and scalable maintenance across the development lifecycle.

3.5. Software Configuration and Production Environment

Software Configuration

The Meeting Booking System adopts a modular, layered configuration to ensure responsive user interaction, secure data handling, and scalable deployment.

- **Frontend Configuration:**

The frontend is developed using Vue.js 3 with TypeScript and Vite, structured as a Single Page Application (SPA). Element Plus ensures consistent UI components, while Vue Router and Axios handle routing and backend communication. ECharts and FullCalendar support visualization and scheduling (Jansen, Vane and Wolff, 2016). This setup enables a responsive and interactive client-side experience.

- **Backend Configuration:**

The backend uses Spring Boot with an MVC structure, supported by MyBatis-Plus for ORM and Druid for connection pooling. Sa-Token and Spring Security Crypto manage authentication, authorization, and secure user session control. The backend ensures secure, scalable, and efficient business processing.

- **Database Configuration:**

MySQL is employed with a 3NF-compliant schema, ensuring data normalization and integrity through foreign key constraints. Navicat assists in schema management and query debugging during development. The database layer guarantees reliable data consistency and scalability.

Production Environment

System	Ubuntu 22.04 64x
Processor	Intel ® Xeon ® Platinum Scalable Processors, 4-core (vCPU), 8 GiB
Public network	bandwidth 5 Mbps
Java OpenJDK	version 17.0.14
MySQL	Version 5.7.27

Our system is deployed on the **AliyunServer** (ecs.e-c1m2.xlarge) with enhanced network performance and dedicated VPC. This ensures high availability, secure isolation, and stable operation for our application.

4. Software testing

We wrote code to test the functionality of the software, mainly focusing on unit and integration tests, as shown in **APPENDIX 11, overview of our test part in IDE**. Acceptance tests were performed by using the web, and all of the tests results are **'pass'**.

4.1. Unit Test

Unit tests verify isolated functionality such as logic correctness and method response, ensuring that each unit of code works accurately. We used the JUnit framework, Mockito, and other simulation tools to emulate the service layer behavior so that the controller methods are tested independently of the Spring Boot lifecycle to ensure that the individual functions and components work as expected before integration. We focus on **User Registration**.

Test Description:Create a simulated user object containing a username and password, then by simulating the behavior of the user's service layer (assuming that the user does not exist and that the save operation is successful), call the registration interface and check that the relevant service methods (query user, save user, and roles) have been called correctly and as expected.

Test Code:

```
@test
void testRegister_Success() {
    // Arrange:Set up test data and mock dependencies
    SysUser newUser = new SysUser();
    newUser.setId(1L); // Set user ID
    newUser.setUsername("testuser"); // Set username
    newUser.setPassword("123456"); // Set password

    // Mock service layer responses to simulate scenarios:
    when(sysUserService.getById(any(Long.class))).thenReturn(null); // User does not exist
    when(sysUserService.save(any(SysUser.class))).thenReturn(true); // User saved successfully
    when(sysUserRoleService.save(any(SysUserRole.class))).thenReturn(true); // User-role saved successfully

    // Act:Execute the method under test
    ResultVo result = sysUserController.register(newUser);

    // Assert: Verify the expected outcomes
    assertEquals(200, result.getCode()); // Check status code
    assertEquals("Registration successful!", result.getMsg()); // Verify success message

    // Verify service layer interactions:
    verify(sysUserService, times(1)).getById(any(Long.class)); // User existence check was performed
    verify(sysUserService, times(1)).save(any(SysUser.class)); // User creation was attempted
    verify(sysUserRoleService, times(1)).save(any(SysUserRole.class)); // Role assignment was attempted
}
```

Figure 14 / APPENDIX 12 : Unit test of user registration

Test Results:

Test Name	CPT202 Unit Test-System User Controller Test				
Test Date	2025.4.28				
Test Number	Test Case	Data (user Id,username,password)	Expected Result	Actual Result	Status
1	Register_Success	1,testUser,123456	code=200, message="Registration successful"	code=200, message="Registration successful"	Pass
2	Register_UserAlreadyExists	1,existingUser,null	code=500, message="The user already exists. Please try a different one"	code=500, message="The user already exists. Please try a different one"	Pass
3	Register_SaveUserFails	1,testUser,password123	code=500, message="Registration failed. Please try again"	code=500, message="Registration failed. Please try again"	Pass
4	Register_SaveUserRoleFails	1,testUser,123456	code=500, message="Registration failed. Please try again"	code=500, message="Registration failed. Please try again"	Pass

Figure 15 : Unit test results

4.2. Integration Test

Integration testing verifies the correctness of interfaces, interactions and co-operation between multiple modules, which includes controller-to-service communication, database interaction and data validation mechanisms.We focus on **new meeting room booking**.

Test Case:New meeting room booking

Test Description:We used the Spring Boot framework and JUnit testing framework, as well as the MockMvc tool to simulate HTTP requests and implement JSON serialization through ObjectMapper to test an interface for booking meeting rooms. First we constructed the request object containing the title, remark and start/end time, sent a POST request to the interface and converted the request body to JSON format; then we verified that the HTTP response status code was as expected and checked the correctness of specific fields in the returned data.

Test Code:

```
@SpringBootTest
@RunWith(MockMvc.class)
public class MeetingBookIntegrationTest {

    @Autowired
    private MockMvc mockMvc;

    @Autowired
    private ObjectMapper objectMapper;

    @Test
    public void testCreateMeetingBook() throws Exception {
        // Arrange: Prepare test data
        MeetingBookRequest request = new MeetingBookRequest();
        request.setTitle("Team Meeting"); // Set meeting title
        request.setRemark("Discuss project updates"); // Add meeting description
        request.setStartTime(LocalDate.parse("2025-04-20T10:00:00")); // Define start time
        request.setEndTime(LocalDate.parse("2025-04-20T11:00:00")); // Define end time

        // Act & Assert: Execute request and validate response
        mockMvc.perform(post("/api/meeting-book") // POST request to endpoint
            .contentType(MediaType.APPLICATION_JSON) // Set JSON content type
            .content(objectMapper.writeValueAsString(request))) // Serialize request body
            .andExpect(status().isOk()) // Verify 200 OK status
            .andExpect(jsonPath("$.id", is(not(empty())))); // Check non-empty response
            // .andExpect(jsonPath("$.roomId", notNullValue())); // Ensure "id" exists in response
            // .andExpect(jsonPath("$.title", is("Team Meeting"))); // Check for title
            // .andExpect(jsonPath("$.remark", is("Discuss project updates"))); // Check for remark
    }
}
```

Figure 16 / APPENDIX 13 : Integration test of new meeting room reservation

Test Results:

Test Name	CPT202 Integration Test-Meeting Book Integration Test				
Test Date	2025.4.28				
Test Number	Test Case	Data	Expected Result	Actual Result	Status
1	CreateMeetingBook	Request body: POST/app/meeting/book Parameters: title="Team Meeting", remark="Discuss project updates", startTime="2025-04-20T10:00", endTime="2025-04-20T11:00"	code=200,Response Body Non Empty (jsonPath("\$.id", not(empty())))	code=200,Response Body Non Empty	Pass
2	GetMeetingBookById	Request: GET/app/meeting/book/1 Assuming that the meeting record with ID 1 exists	code=200,Response Body Non Empty	code=200,Response Body Non Empty	Pass
3	UpdateMeetingBook	Request body: title="Updated Meeting", remark="Discuss project updates", startTime="2025-04-20T10:00:00", endTime="2025-04-20T11:00:00". Send PUT request to/app/meeting/book/1	code=200,Response Body Non Empty	code=200,Response Body Non Empty	Pass
4	DeleteMeetingBook	Send a DELETE request to/app/meeting/book/1, assuming that a meeting record	code=200,Response Body Non Empty	code=200,Response Body Non Empty	Pass
5	GetAllMeetingBooks	Send GET request to/app/meeting/book	code=200,Response Body Non Empty	code=200,Response Body Non Empty	Pass

Figure 17 : Integration test results

4.3. Acceptance Test

Acceptance testing is to verify that the system under the software meets the business requirements and can function properly in real applications and be accepted by users. Our acceptance testing includes business process testing, compatibility and user experience testing, security and performance testing, and accessibility testing.

Test Name	CPT202 Acceptance Test			
Test Date	2025.4.23			
Test Number	Test Input	Expected Result	Actual Result	Result
1	Select correct range of beginning and ending date	Display correct result of selected transactions with volume and transaction	Actual result displayed	Pass
2	Select incorrect range of date	Display error message 'Invalid setting, please check it!' in red	Error message displayed	Pass
3	Press 'generate' button with expected time range	Generate two figures of statistical information (order volume and total revenue)	Figures generated as expected	Pass

Figure 18 : Test Case ID:PBI-032 Users cannot book for more than 8 hours

In addition, the user interface (UI) should enable users to use the web pages and complete meeting room bookings efficiently, easily and pleasantly. Acceptance testing is not just about checking that the UI elements 'look right', but more importantly verifying that the UI 'works right'. This is an example of user and meeting room visual charts.

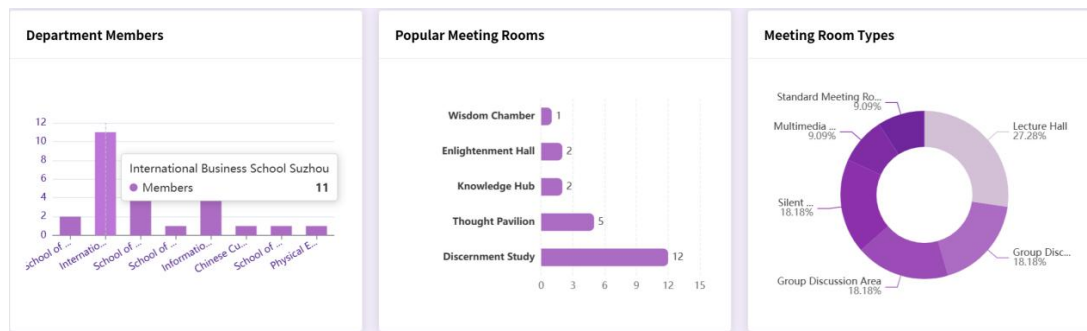


Figure 19 : User and meeting room usage dashboard

5. Conclusion and future work

So far, our group has transformed the project requirements into a web page that can be operated on. We adopt the MVC architecture to implement a decoupled and maintainable framework. The frontend and backend are built using Vue.js and Spring Boot, providing a responsive user experience and powerful server-side processing. Advanced database design ensures data integrity and consistency, while software modules customized for different user roles support room booking, availability viewing, maintenance requests, and resource management operations. Our website can already meet all the needs of the project.

Although we have achieved some success, there are still some imperfections in the implementation process of the project. Firstly, our website is currently only compatible with computers, and its compatibility on devices such as smartphones and iPads still needs to be improved. Secondly, in our design, professional operators are required to approve and control the booking of meeting rooms, which requires administrators to be online in real-time and respond to user needs in a timely manner. We hope that this feature can be automated in the future. Finally, as the project progresses, if the requirements change, our role-based module partitioning may make it difficult to add new features, which requires some adjustments to the initial plan to incorporate these changes into the design and implementation process.

Looking ahead, our group plans to continue improving and enhancing the online meeting booking system. Next, we will focus on optimizing system performance, improving its response speed and stability under high concurrency conditions, including further optimizing databases, code structures, and server processing capabilities. At the same time, we hope to better understand everyone's feelings and needs during the operation process by collecting user feedback. Based on this feedback, we will continue to improve system functionality and user interface design, such as finding more suitable optimization directions for users through A/B testing, in order to enhance overall user experience and satisfaction. In addition, we also plan to explore adding some new features, such as integration with video conferencing platforms, providing more detailed resource usage analysis reports, etc., to further enrich the application scenarios of the system and make meeting room management more intelligent and efficient. Through continuous improvement and efforts, we hope to make this meeting booking system truly meet the actual needs of the school, improve resource management level, and provide reliable reference for the construction of similar systems in the future.

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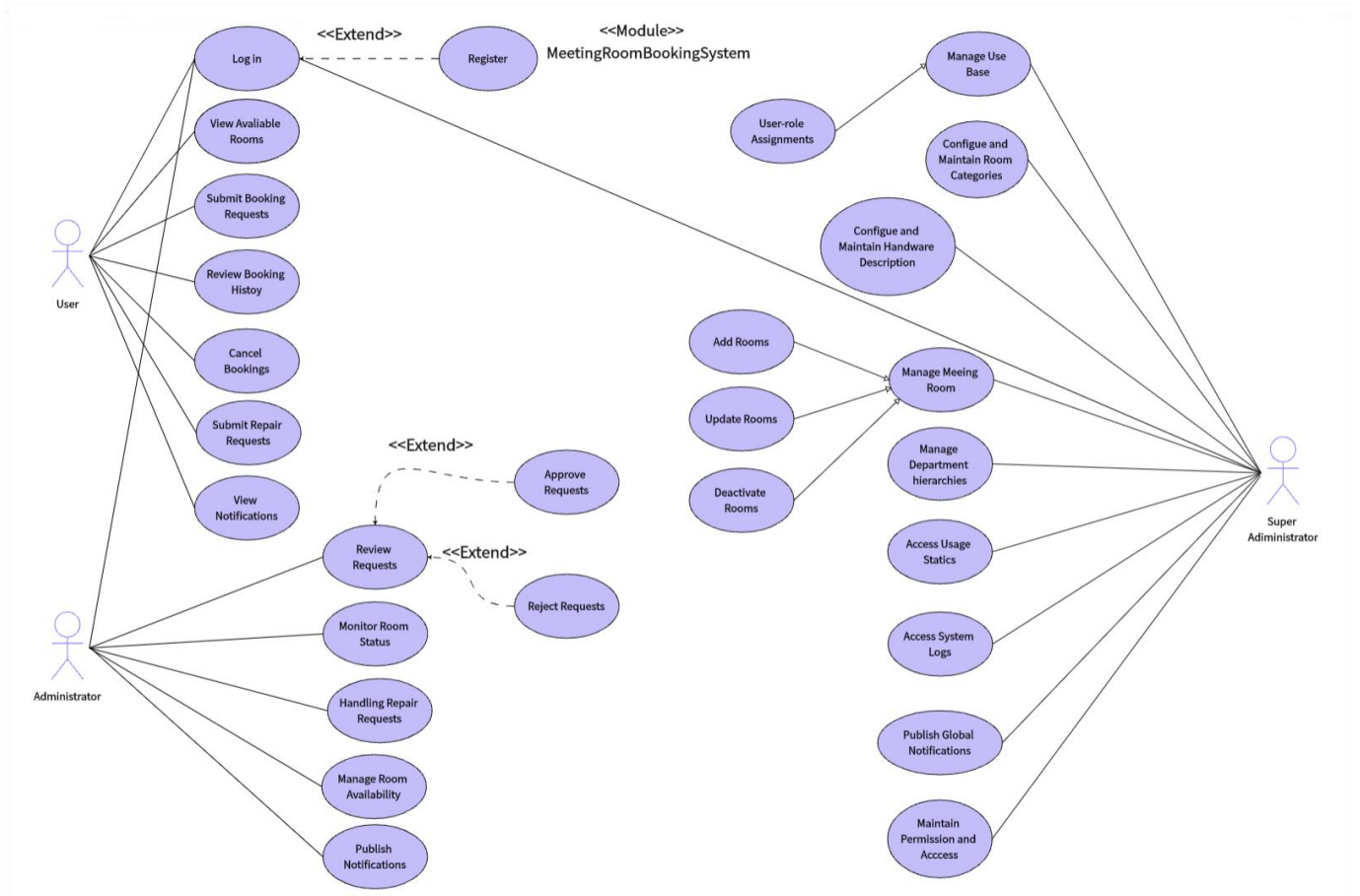
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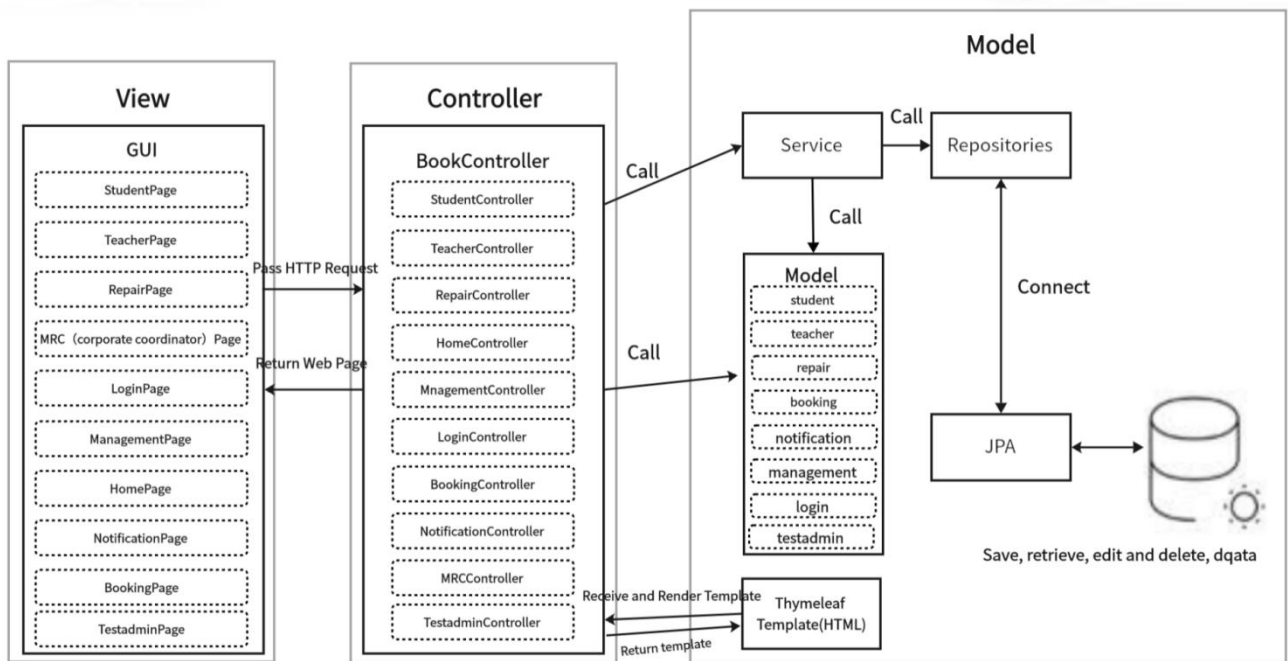
7. Appendix

If the figures in the main text are not clear, the following appendix provides high-definition versions.

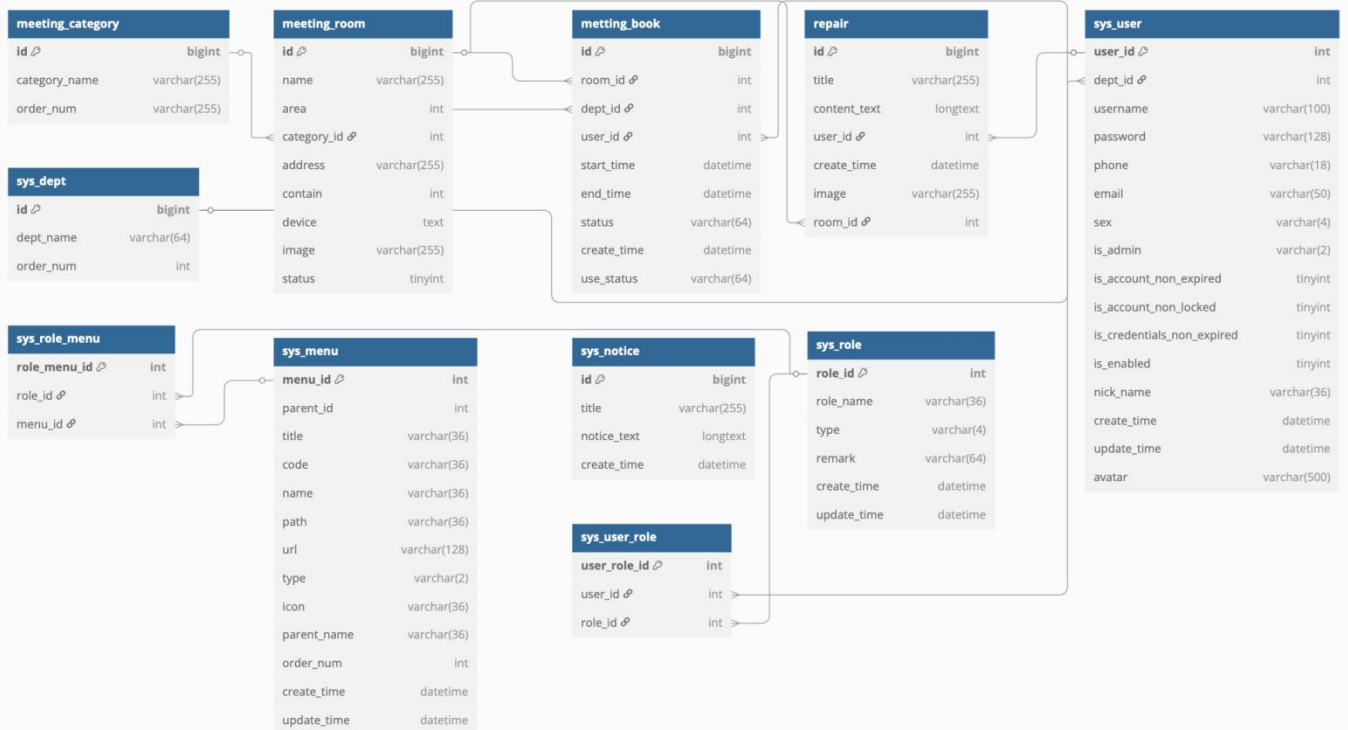
APPENDIX 1 - Figure 1 : Use Case Diagram of the System



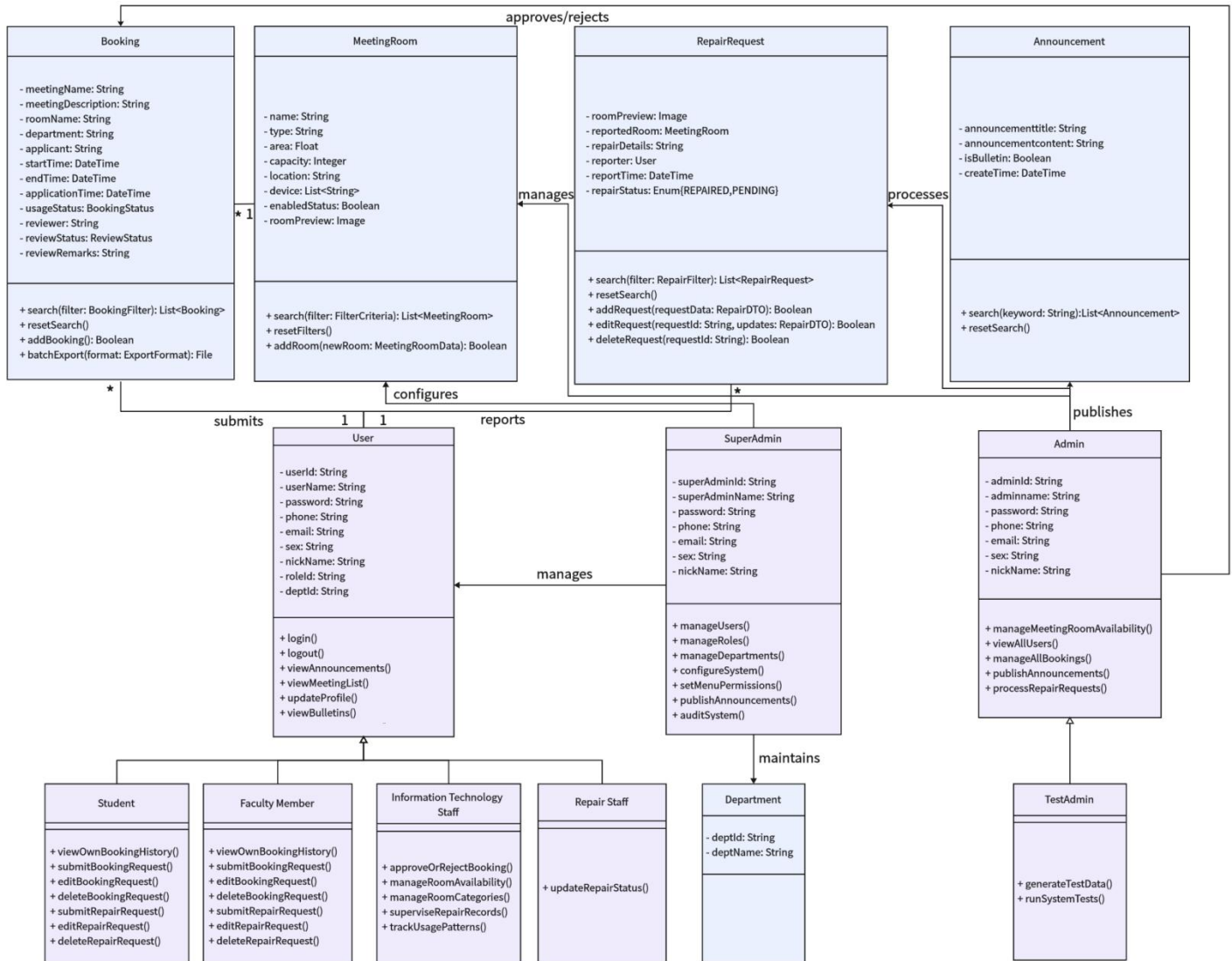
APPENDIX 2 - Figure 2 : MVC architecture



APPENDIX 3 - Figure 3 : ER Diagram



Appendix 4 - Figure 4 : Software Modules Design

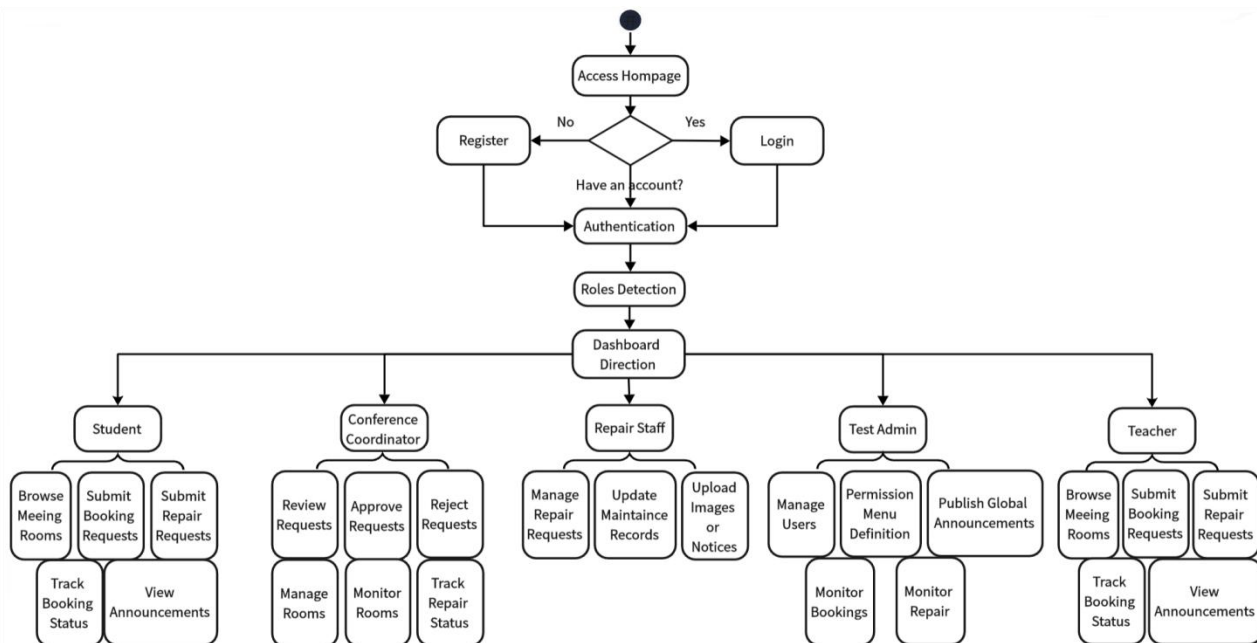


APPENDIX 5 - Figure 6 : Role-Based Menu Permission Code

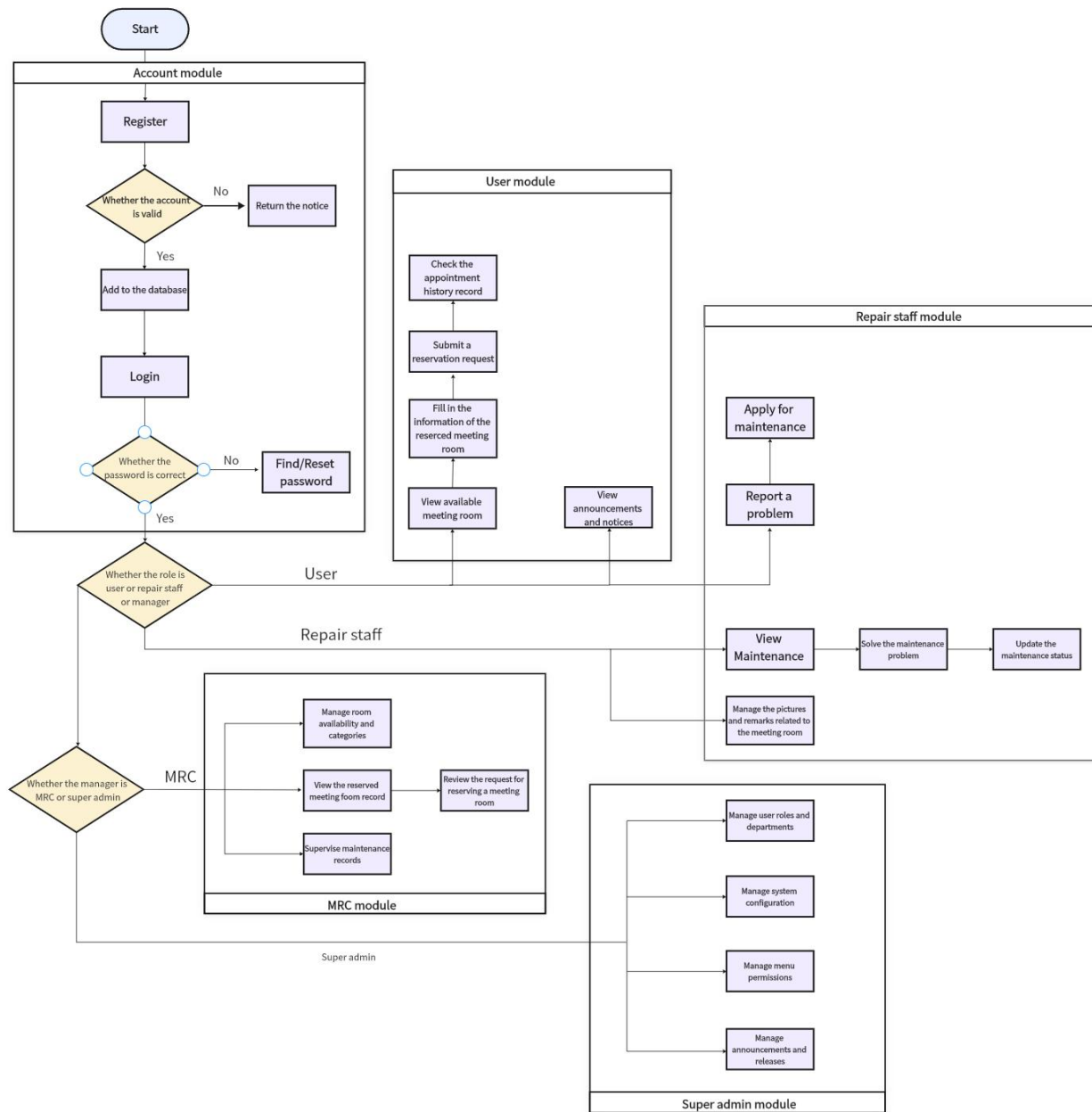
```
// Initialize an empty menu list
List<SysMenu> menuList = null;
// Check if the user is a super administrator
if (StringUtils.isEmpty(user.getIsAdmin()) && "1".equals(user.getIsAdmin())) {
    // Super administrator, directly query all menus
    menuList = sysMenuService.list();
} else {
    // For regular users: get menus assigned to their user ID
    menuList = sysMenuService.getMenuByUserId(user.getUserId());
}
// Validate if the user has any valid menu permissions
if (menuList.stream().filter(Objects::nonNull).collect(Collectors.toList()).isEmpty()) {
    return ResultUtils.error("The user's role has not been assigned menu permissions. Please log in with an administrator account to assign menus.");
}

// Get the 'code' field from the menu table
List<String> collect = Optional.ofNullable(menuList).orElse(new ArrayList<>())// Handle potential null list, use empty list if null
    .stream()
    .filter(item -> item != null && StringUtils.isEmpty(item.getCode())) // Filter items with empty codes
    .map(SysMenu::getCode)// Extract code field
    .collect(Collectors.toList());// Collect to string list
```

APPENDIX 6 - Figure 7 : Process Flow



APPENDIX 7 - Figure 8 : Role login division flow chart



APPENDIX 8-Figure 9 : Login Authentication flow chart code (1)

```
//Login
@PostMapping("/login")
public ResultVo login(HttpServletRequest request, @RequestBody LoginParm parm) {
    //Get the data passed from the frontendcode
    String code = parm.getCode();
    //Acquiresession
    HttpSession session = request.getSession();
    System.out.println("Session ID: " + session.getId());
    System.out.println("Session Creation Time: " + new Date(session.getCreationTime()));
    System.out.println("Session Last Access Time: " + new Date(session.getLastAccessedTime()));

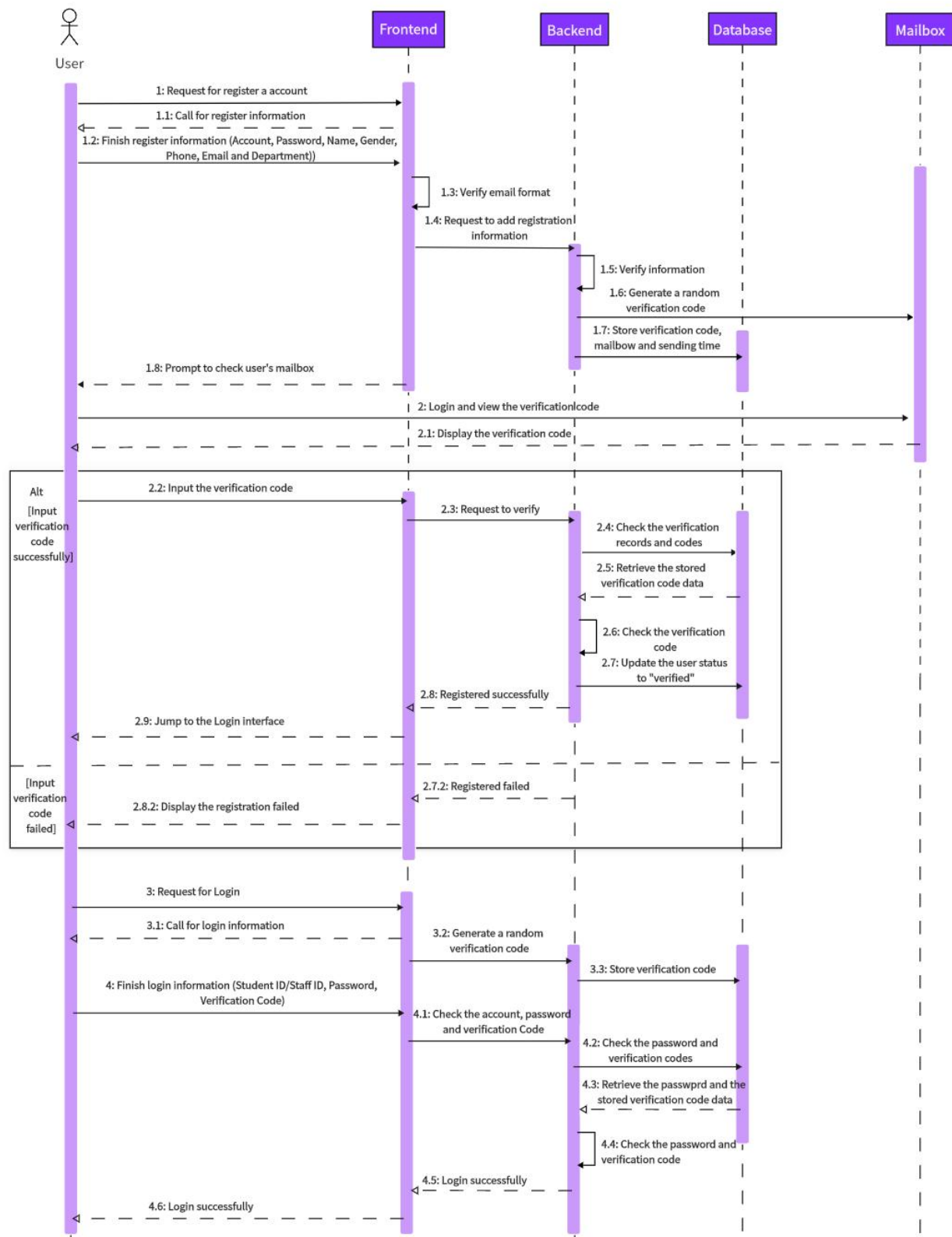
    //Acquiresessioninsidecode
    String code1 = (String) session.getAttribute("code");
    System.out.println("Code in session: " + code1);

    // if (StringUtils.isEmpty(code1)) {
    //     return ResultUtils.error("Verification code expired!");
    // }
    //Determine the incoming frontend parameterscodeandsessionIs the content inside equal?
    // if (!code1.equals(code)) {
    //     return ResultUtils.error("Verification code is incorrect!");
    // }
```

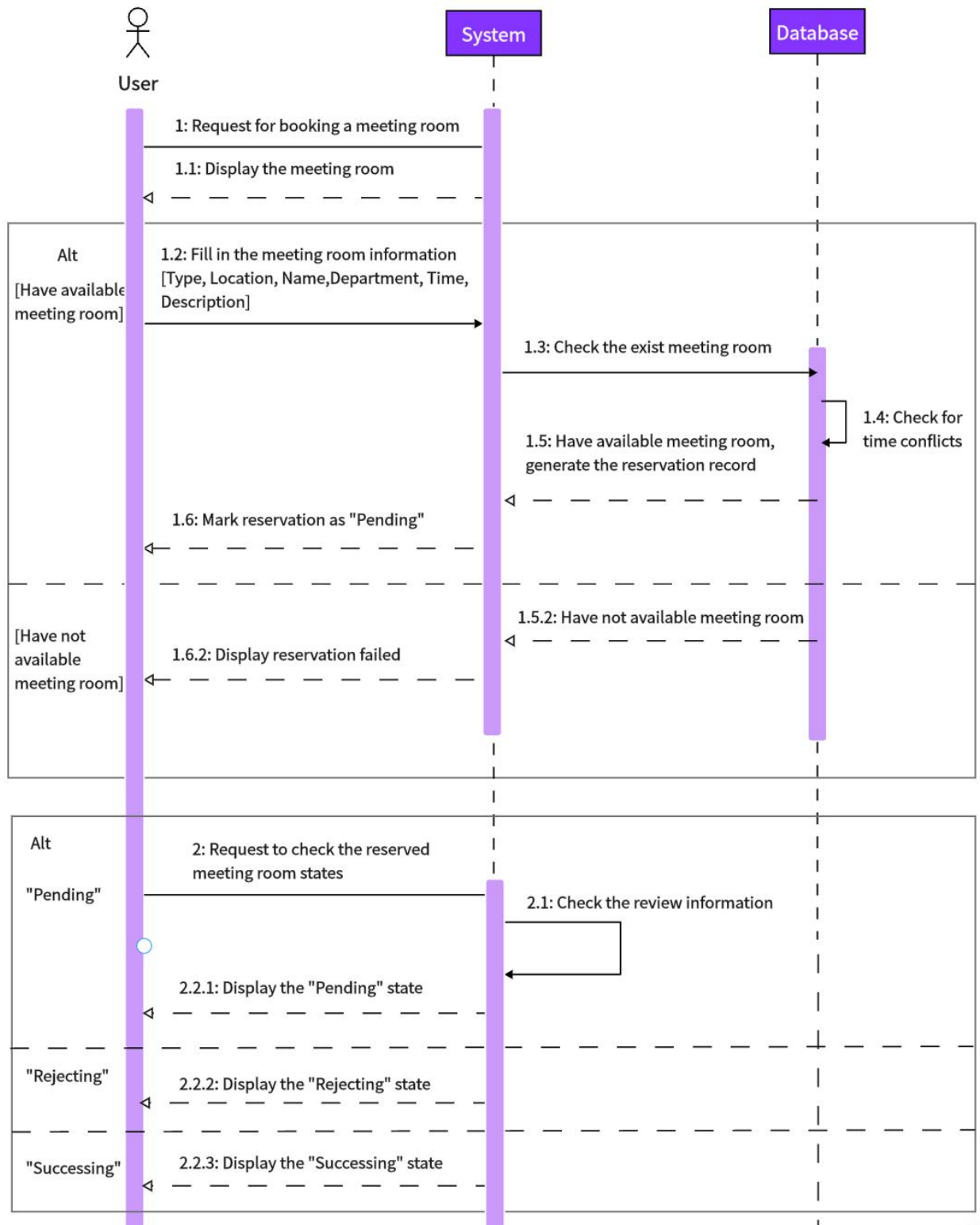
APPENDIX 8-Figure 10 : Login Authentication flow chart code (2)

```
QueryWrapper<SysUser> query = new QueryWrapper<>();
query.lambda().eq(SysUser::getUsername, parm.getUsername());
SysUser user = sysUserService.getOne(query);
if(null == user){
    return ResultUtils.error("Username error!");
}
if (!passwordEncoder.matches(parm.getPassword(), user.getPassword())) {
    return ResultUtils.error("Wrong password!");
}
if (!user.isEnabled()) {
    return ResultUtils.error("Account has been banned, please contact the system administrator!");
}
StpUtil.login(user.getUserId(), "PC");
//Return user information andtoken
LoginVo vo = new LoginVo();
vo.setUserId(user.getUserId());
vo.setNickName(user.getNickName());
//Generatetoken
// First2step, Acquire Token Related parameters
SaTokenInfo tokenInfo = StpUtil.getTokenInfo();
vo.setToken(tokenInfo.getTokenValue());
return ResultUtils.success("Login successful", vo);
}
```

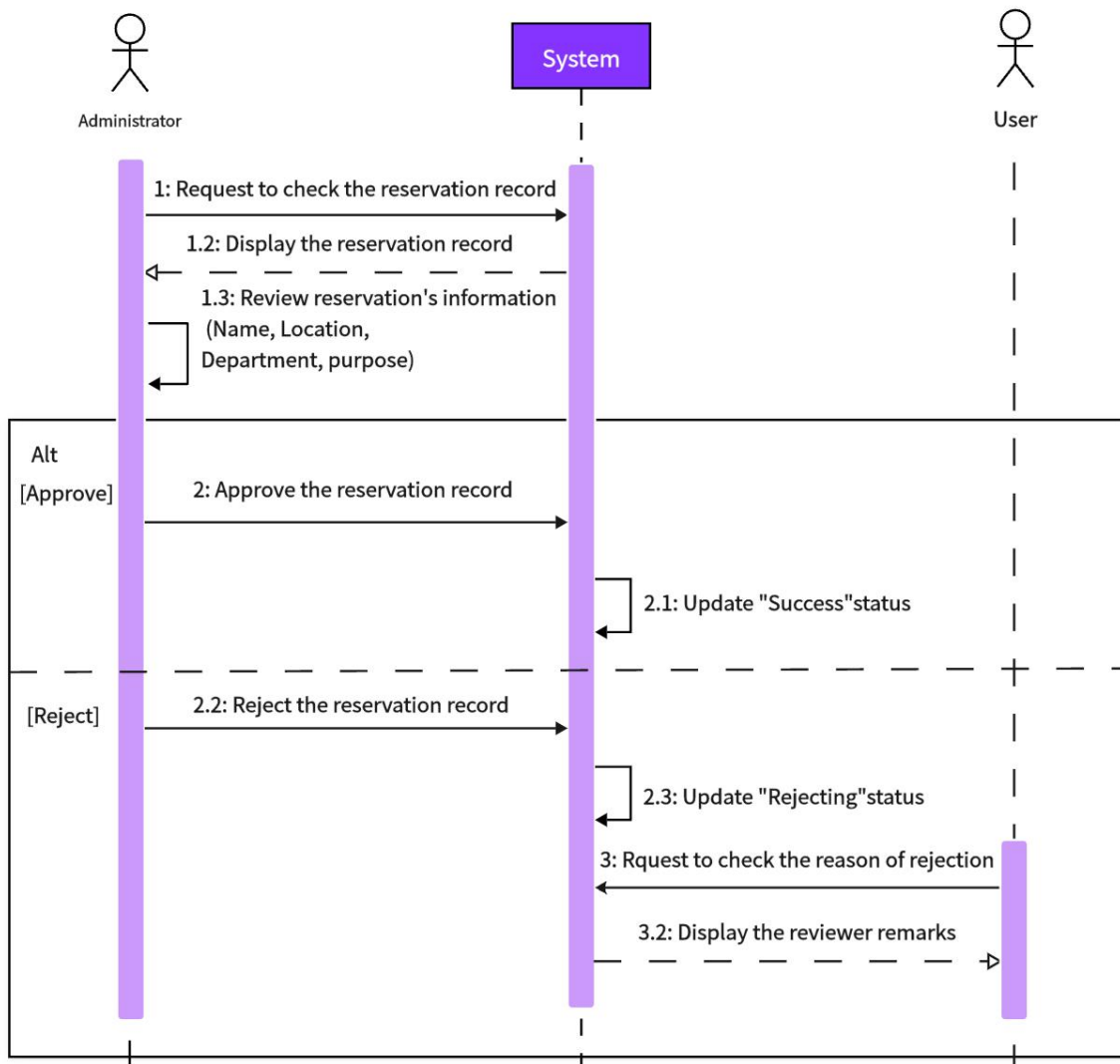
APPENDIX 9 - Figure 11 : User registration and login



APPENDIX 9 - Figure 12 : User reserved meeting room



APPENDIX 9 - Figure 13 : Administrator reviews meeting room booking records



APPENDIX 10 : Java directory structure

```

|---/src/main/java/com/group17/CPT202_Group17_Backend
|   |---/config: Configuration class storage location
|   |---/controller: Control layer
|   |---/service: Business code layer
|   |---/entity: Storing database entity classes
|   |---/mapper: Interaction layer with database
|
|---/src/main/resources: Stores resource files, such as static resource files, configuration
files, page files, etc.
|   |---/mapper: XML files that store the MyBatis operation database
|   |---/static: Stores static resources such as CSS styles, images, etc.
|   |---/templates: Stores template files such as Thymeleaf template files
|
|---/src/test/java/com/group17: Unit test class and integration test class Java files
    
```

APPENDIX 11 : Overview of our test part in IDE

3/3 19.7s

web 80.1s

- ✓ { } com.group17.web.auth 13.0s
 - ✓ ⚙️ AuthControllerIntegrationTest 5.6s
 - ✓ testSendCode_shouldReturnSuccess() 4.0s
 - ✓ testVerifyCode_shouldReturnSuccessIfCorrect() 1.6s
 - ✓ ⚙️ AuthControllerTest 7.4s
 - ✓ testSendCode_ShouldCallSendEmailInternally() 2.0s
 - ✓ testSendCode_EmailContentCorrect() 3.1s
 - ✓ testSendCode_EmailSendFails() 2.2s
- ✓ { } com.group17.web.home.controller 32.3s
 - ✓ ⚙️ HomeControllerIntegrationTest 2.6s
 - ✓ testGetTotalIntegration() 2.6s
 - ✓ testGetEchartsIntegration() 18ms
 - ✓ testGetNoticesIntegration() 14ms
 - ✓ ⚙️ HomeControllerTest 29.7s
 - ✓ testGetTotal() 3.1s
 - ✓ testGetEcharts() 6.0s
 - ✓ testGetMeetings() 5.5s
 - ✓ testGetNotices() 4.8s
 - ✓ testGetLastMeeting() 5.1s
 - ✓ testGetCategoryBook() 5.2s
- ✓ { } com.group17.web.meeting_book.controller 4.5s
 - ✓ ⚙️ MeetingBookControllerTest 1.7s
 - ✓ testApply_Success() 19ms
 - ✓ testApply_Failure() 5.0ms
 - ✓ testApply_NullMeetingBook() 1.7s
 - ✓ ⚙️ MeetingBookIntegrationTest 2.8s
 - ✓ testCreateMeetingBook() 16ms
 - ✓ testGetMeetingBookById() 2.6s
 - ✓ testUpdateMeetingBook() 169ms
 - ✓ testDeleteMeetingBook() 14ms
 - ✓ testGetAllMeetingBooks() 16ms

- ✓ { } com.group17.web.meeting_room.controller 431ms
 - ✓ ⚙️ MeetingRoomControllerIntegrationTest 46ms
 - ✓ testGetRoomListIntegration() 21ms
 - ✓ testSearchMeetingRoomIntegration() 13ms
 - ✓ testSelectListIntegration() 12ms
 - ✓ ⚙️ MeetingRoomControllerTest 385ms
 - ✓ testGetRoomList_NonEmptyList() 279ms
 - ✓ testSelectList_ReturnsDropDownOptions() 58ms
 - ✓ testDeleteRoom_Successful() 48ms
- ✓ { } com.group17.web.sys_role.controller 342ms
 - ✓ ⚙️ SysRoleControllerIntegrationTest 47ms
 - ✓ testAddRole_IntegrationSuccess() 29ms
 - ✓ testGetRoleList_Success() 8.0ms
 - ✓ testGetRoleSelectList_Success() 10ms
 - ✓ ⚙️ SysRoleControllerTest 295ms
 - ✓ testAdd_Success() 244ms
 - ✓ testGetList_WithRoleName() 43ms
 - ✓ testSelectList_ReturnsFormattedDropDown() 8.0ms
- ✓ { } com.group17.web.sys_user.controller 3.5s
 - ✓ ⚙️ SysUserControllerIntegrationTest 3.2s
 - ✓ testGetInfo_shouldReturnUserInfoAndPermissions() 51ms
 - ✓ testImageCode_shouldReturnBase64Captcha() 1.4s
 - ✓ testRegister_shouldCreateNewUser() 1.8s
 - ✓ ⚙️ SysUserControllerTest 280ms
 - ✓ testRegister_Success() 11ms
 - ✓ testRegister_UserAlreadyExists() 9.0ms
 - ✓ testRegister_SaveUserFails() 248ms
 - ✓ testRegister_SaveUserRoleFails() 12ms

APPENDIX 12 - Figure 14 : Unit test of user registration

```
@Test
void testRegister_Success() {
    // Arrange:Set up test data and mock dependencies
    SysUser newUser = new SysUser();
    newUser.setUserId(1L); // Set user ID
    newUser.setUsername("testUser"); // Set username
    newUser.setPassword("123456"); // Set password

    // Mock service layer responses to simulate scenarios:
    when(sysUserService.getById(any(Long.class))).thenReturn(null); // User does not exist
    when(sysUserService.save(any(SysUser.class))).thenReturn(true); // User saved successfully
    when(sysUserRoleService.save(any(SysUserRole.class))).thenReturn(true); // User-role saved successfully

    // Act:Execute the method under test
    ResultVo result = sysUserController.register(newUser);

    // Assert: Verify the expected outcomes
    assertEquals(200, result.getCode()); // Check status code
    assertEquals("Registration successful!", result.getMsg()); // Verify success message

    // Verify service layer interactions:
    verify(sysUserService, times(1)).getById(any(Long.class)); //User existence check was performed
    verify(sysUserService, times(1)).save(any(SysUser.class)); //User creation was attempted
    verify(sysUserRoleService, times(1)).save(any(SysUserRole.class)); //Role assignment was attempted
}
```

APPENDIX 13 - Figure 16 : Integration test of new meeting room reservation

```
@SpringBootTest
@AutoConfigureMockMvc
public class MeetingBookIntegrationTest {

    @Autowired
    private MockMvc mockMvc;

    @Autowired
    private ObjectMapper objectMapper;

    @Test
    public void testCreateMeetingBook() throws Exception {
        // Arrange: Prepare test data
        MeetingBookRequest request = new MeetingBookRequest();
        request.setTitle("Team Meeting"); // Set meeting title
        request.setRemark("Discuss project updates"); // Add meeting description
        request.setStartTime(LocalDateTime.parse("2025-04-20T10:00:00")); // Define start time
        request.setEndTime(LocalDateTime.parse("2025-04-20T11:00:00")); // Define end time

        // Act & Assert: Execute request and validate response
        mockMvc.perform(post("/api/meeting-book") // POST request to endpoint
            .contentType(MediaType.APPLICATION_JSON) // Set JSON content type
            .content(objectMapper.writeValueAsString(request))) // Serialize request body
            .andExpect(status().isOk()) // Verify 200 OK status
            .andExpect(jsonPath("$", is(not(empty())))); // Check non-empty response
        // .andExpect(jsonPath("$.roomId", notNullValue())) // Ensure "id" exists in response
        // .andExpect(jsonPath("$.title", is("Team Meeting"))) // Check for title
        // .andExpect(jsonPath("$.remark", is("Discuss project updates"))); // Check for remark
    }
}
```

APPENDIX 14 : Sprint Process Report

Sprint Progress Report						
Project Name	An online meeting booking system					
Team Number	17					
Team Name	Group17					
Team Leader	Xiangyi.Zhu					
TA	Tong.Zou					
Week	Sprint	PBI	Owener	Start Date	Expected CompletionDate	Actual Completion Date
1	1	Write requirement documentation.	Xiangyi Zhu	2/17/2025	2/18/2025	2/18/2025
	1	Choose technology stack	Xiangyi Zhu	2/17/2025	2/18/2025	2/18/2025
	1	Initialize project repository	Xinrong Shi	2/17/2025	2/18/2025	2/18/2025
	1	Design user registration page	Hanqiu Zhang	2/19/2025	2/21/2025	2/21/2025
	1	Develop registration API	Hanqiu Zhang	2/19/2025	2/21/2025	2/21/2025
	1	Implement email verification.	Xiangyi Zhu	2/19/2025	2/21/2025	2/21/2025
	1	Design login page	Xinyuan Zhang	2/19/2025	2/21/2025	2/21/2025
	1	Develop login API	Xiangyi Zhu	2/22/2025	2/23/2025	2/23/2025
	1	Create user profile editing page	Nan Lin	2/22/2025	2/23/2025	2/23/2025
	1	Configure admin menu permissions	Nan Lin	2/22/2025	2/23/2025	2/23/2025
2	2	Load dynamic menus based on roles	Zimo Zheng	2/24/2025	2/27/2025	2/27/2025
	2	Develop permission middleware	Xinrong Shi	2/24/2025	2/27/2025	2/27/2025
	2	Design room category management	Nan Lin	2/24/2025	2/27/2025	2/27/2025
	2	Develop room CRUD APIs	Zimo Zheng	2/24/2025	2/27/2025	2/27/2025
	2	Add room status labels (available/disabled)	Zwei Zhou	2/24/2025	2/27/2025	2/27/2025
	2	Design room search/filter UI	Zwei Zhou	2/24/2025	2/27/2025	2/27/2025
	2	Implement time conflict detection (back-end)	Xiangyi Zhu	28/2/2025	3/2/2025	3/2/2025
	2	Develop booking approval workflow	Jiayi Wang	28/2/2025	3/2/2025	3/2/2025
	2	Create user booking history page	Jiayi Wang	28/2/2025	3/2/2025	3/2/2025
	2	Encrypt passwords	Xinrong Shi	28/2/2025	3/2/2025	3/2/2025
3	3	Prevent SQL injection	Nan Lin	3/3/2025	3/9/2025	3/9/2025
	3	Cache sessions with Redis	Xiangyi Zhu	3/3/2025	3/9/2025	3/9/2025
	3	Optimize database indexes	Hanqiu Zhang	3/3/2025	3/9/2025	3/9/2025
	3	Add email notifications	Xiangyi Zhu	3/3/2025	3/9/2025	3/9/2025
	3	Log admin operations	Jiayi Wang	3/3/2025	3/9/2025	3/9/2025
	3	Build data dashboards	Xinrong Shi	3/3/2025	3/9/2025	3/9/2025
	3	Write unit tests (JUnit)	Xinrong Shi	3/3/2025	3/9/2025	3/9/2025
	3	Test APIs with Postman	Zwei Zhou	3/3/2025	3/9/2025	3/9/2025
	3	Dockerize backend/frontend	Jiayi Wang	3/3/2025	3/9/2025	3/9/2025
	3	Configure Nginx and HTTPS	Hanqiu Zhang	3/3/2025	3/9/2025	3/9/2025
4	3	Set up CI/CD pipeline	Hanqiu Zhang	3/3/2025	3/9/2025	3/9/2025
	3	Monitor server logs (ELK Stack)	Nan Lin	3/3/2025	3/9/2025	3/9/2025
	4	Collect user feedback	Zimo Zheng	3/10/2025	3/12/2025	3/12/2025
	4	Plan version updates (V1.0 → V1.2)	Xiangyi Zhu	3/10/2025	3/12/2025	3/12/2025
	4	Design password reset page	Zimo Zheng	3/10/2025	3/12/2025	3/12/2025
	4	Develop password reset API	Xiangyi Zhu	3/10/2025	3/12/2025	3/12/2025
	4	Create admin dashboard layout	Zimo Zheng	3/10/2025	3/12/2025	3/12/2025
	4	Add room image upload functionality	Xinyuan Zhang	3/13/2025	3/16/2025	3/16/2025
	4	Implement booking cancellation logic	Xinyuan Zhang	3/13/2025	3/16/2025	3/16/2025
	4	Validate user input (front-end/back-end)	Zimo Zheng	3/13/2025	3/16/2025	3/16/2025
5	4	Handle token expiration (JWT)	Zimo Zheng	3/13/2025	3/16/2025	3/16/2025
	4	Generate monthly usage reports	Xinyuan Zhang	3/13/2025	3/16/2025	3/16/2025
	5	Integrate third-party calendars	Zwei Zhou	3/17/2025	3/20/2025	3/20/2025
	5	Monitor system performance	Xiaoya Li	3/17/2025	3/20/2025	3/20/2025
	5	Backup database regularly	Xinyuan Zhang	3/17/2025	3/20/2025	3/20/2025
	5	Implement rate limiting	Jiayi Wang	3/17/2025	3/20/2025	3/20/2025
	5	Design error handling pages	Xinyuan Zhang	3/17/2025	3/20/2025	3/20/2025
	5	Test cross-browser compatibility	Xinrong Shi	3/17/2025	3/20/2025	3/20/2025
	5	Optimize mobile responsiveness	Zwei Zhou	3/21/2025	3/23/2025	3/23/2025
	5	Conduct load testing	Xiaoya Li	3/21/2025	3/23/2025	3/23/2025
6	6	Document API endpoints	Zwei Zhou	3/24/2025	3/30/2025	3/30/2025
	6	Localize UI for multilingual support	Jiayi Wang	3/24/2025	3/30/2025	3/30/2025
	6	Add keyboard navigation support	Hanqiu Zhang	3/24/2025	3/30/2025	3/30/2025
	6	Implement dark mode toggle	Xiaoya Li	3/24/2025	3/30/2025	3/30/2025
	6	Audit code for security vulnerabilities	Xinrong Shi	3/24/2025	3/30/2025	3/30/2025
	6	Update dependencies regularly	Xiaoya Li	3/24/2025	3/30/2025	3/30/2025
7	6	Train admin users	Xiaoya Li	3/24/2025	3/30/2025	3/30/2025
	7	Create user guides/FAQs	Nan Lin	3/31/2025	4/6/2025	4/6/2025
	7	Schedule periodic system reviews	Nan Lin	3/31/2025	4/6/2025	4/6/2025

APPENDIX 15 (1) : Database Table Structure Documentation

Database Table Structure Documentation

Database Name: meeting-room-system

Version: 1.0.0

Description: Database Design Documentation

Table:meeting_category (Meeting Room Type)

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	id	bigint	20	0	N	Y		Serial number
2	category_name	varchar	255	0	Y	N		Category name
3	order_num	varchar	255	0	Y	N		Category order No.

Table:meeting_room (Meeting Room Table)

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	id	bigint	20	0	N	Y		Serial number
2	name	varchar	255	0	Y	N		Meeting room name
3	area	int	10	0	Y	N		Area
4	category_id	int	10	0	Y	N		Meeting room category
5	address	varchar	255	0	Y	N		Address
6	contain	int	10	0	Y	N		Capacity
7	device	text	65535	0	Y	N		Device
8	image	varchar	255	0	Y	N		Meeting room images
9	status	tinyint	0	0	Y	N		Enabled status

Table:meeting_book (Appointment Management)

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	id	bigint	20	0	N	Y		Serial number
2	room_id	int	10	0	Y	N		Meeting room name
3	dept_id	int	10	0	Y	N		User department
4	user_id	int	10	0	Y	N		User id

APPENDIX 15 (2) : Database Table Structure Documentation

5	start_time	datetime	19	0	Y	N		Start time
6	end_time	datetime	19	0	Y	N		End time
7	status	varchar	64	0	Y	N		Audit status
8	create_time	datetime	19	0	Y	N		Create time
9	use_status	varchar	64	0	Y	N		Use status

Table : repair (Repair Management)

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	id	bigint	20	0	N	Y		Serial number
2	title	varchar	255	0	Y	N		Title
3	content_text	longtext	2147483647	0	Y	N		Repair details
4	user_id	int	10	0	Y	N		Person who reports a repair
5	create_time	datetime	19	0	Y	N		Time to report for repair
6	image	varchar	255	0	Y	N		Images of reported repairs
7	room_id	int	10	0	Y	N		Meeting room reported for repairing

Table: sys_dept (Department Table)

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	id	bigint	20	0	N	Y		Serial number
2	dept_name	varchar	64	0	Y	N		Department name
3	order_num	int	10	0	Y	N		Department number

Table : sys_menu

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	menu_id	int	10	0	N	Y		Menu id
2	parent_id	int	10	0	Y	N		Parent id
3	title	varchar	36	0	Y	N		Menu title
4	code	varchar	36	0	Y	N		Permission field
5	name	varchar	36	0	Y	N		Route name
6	path	varchar	36	0	Y	N		Routing address

APPENDIX 15 (3) : Database Table Structure Documentation

7	url	varchar	128	0	Y	N		Component path
8	type	varchar	2	0	Y	N		Menu type 0:Catalogue 1:Menu 2:Button
9	icon	varchar	36	0	Y	N		Menu icon
10	parent_name	varchar	36	0	Y	N		Parent menu name
11	order_num	int	10	0	Y	N		Serial number
12	create_time	datetime	19	0	Y	N		Create time
13	update_time	datetime	19	0	Y	N		Update time

Table: **sys_notice** (Announcement Table)

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	id	bigint	20	0	N	Y		Serial number
2	title	varchar	255	0	Y	N		Announcement title
3	notice_text	longtext	2147483647	0	Y	N		Announcement
4	create_time	datetime	19	0	Y	N		Create time

Table: **sys_role**

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	role_id	int	10	0	N	Y		Role id
2	role_name	varchar	36	0	Y	N		Role name
3	type	varchar	4	0	Y	N		Type
4	remark	varchar	64	0	Y	N		Description
5	create_time	datetime	19	0	Y	N		Create time
6	update_time	datetime	19	0	Y	N		Update time

Table : **sys_role_menu**

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	role_menu_id	int	10	0	N	Y		Primary key
2	role_id	int	10	0	Y	N		Role id
3	menu_id	int	10	0	Y	N		Menu id

APPENDIX 15 (4) : Database Table Structure Documentation

Table : sys_user

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	user_id	int	10	0	N	Y		User id
2	dept_id	int	10	0	Y	N		Department id
3	username	varchar	100	0	Y	N		Login account
4	password	varchar	128	0	Y	N		Login password
5	phone	varchar	18	0	Y	N		Telephone number
6	email	varchar	50	0	Y	N		Email address
7	sex	varchar	4	0	Y	N		Gender
8	is_admin	varchar	2	0	Y	N		Whether super administrator 1: Yes 0: No
9	is_account_non_expired	tinyint	4	0	Y	N		Whether the account is expired 1 : Not expired 0 : Expired
10	is_account_non_locked	tinyint	4	0	Y	N		Whether the account is locked 1 : unlocked 0 : locked
11	is_credentials_non_expired	tinyint	4	0	Y	N		Whether the password has expired 1 : not expired 0 : expired
12	is_enabled	tinyint	4	0	Y	N		Whether the account is available 1 : available 0 : removes the user
13	nick_name	varchar	36	0	Y	N		Nick name
14	create_time	datetime	19	0	Y	N		Create time

APPENDIX 15 (5) : Database Table Structure Documentation

15	update_time	datetime	19	0	Y	N		Update time
16	avatar	varchar	500	0	Y	N		Avatar path

Table : sys_user_role

No.	Field Name	Data Type	Length	Decimal	Nullable	Primary Key	Default	Description
1	user_role_id	int	10	0	N	Y		Primary key
2	user_id	int	10	0	Y	N		User id
3	role_id	int	10	0	Y	N		Role id

APPENDIX 16 (1) : Schematic diagram of some key data tables of Navicat--meeting_book

id # bigint	room_id # int	dept_id # int	user_id # int	start_time datetime	end_time datetime	status varchar	create_time datetime	use_status varchar(64)	use_desc varchar(255)	apply_desc varchar(255)	process_user varchar(255)	meeting_name varchar(255)
17	1	4	14	2025-03-18 23:00:00	2025-03-19 00:00:00	3	2025-03-27 17:46:33	4	Group project discussion	Approved	Administrator	Startup Brainstorming
19	5	4	14	2025-03-27 22:00:00	2025-03-28 00:00:00	1	2025-03-27 17:50:46	1	Startup competition preparation	Approved	Conference Coordinator 1	Business Pitch Rehearsal
20	2	4	10	2025-03-06 04:00:00	2025-03-06 06:00:00	1	2025-03-27 19:59:11	1	Group 17 Weekly Discussion	Approved	Conference Coordinator 1	Group Assignment Discussion
21	3	4	10	2025-03-13 04:00:00	2025-03-13 06:00:00	1	2025-03-27 20:01:35	1	Group 17 Weekly Discussion	Approved	Conference Coordinator 1	Group Assignment Discussion
22	4	1	18	2025-03-26 22:00:00	2025-03-27 00:00:00	1	2025-03-27 21:01:16	1	Extracurricular math learning	Approved	Conference Coordinator 2	Math Club Workshop 1
23	4	6	18	2025-04-02 22:00:00	2025-04-03 00:00:00	1	2025-03-27 21:02:40	1	Extracurricular math learning	Approved	Conference Coordinator 2	Math Club Workshop 2
24	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 21:12:29	0	Self-directed learning	(Null)	(Null)	Self-directed learning
25	10	9	16	2025-04-22 23:00:00	2025-04-22 01:00:00	0	2025-04-21 21:13:46	0	Self-directed learning	(Null)	(Null)	Self-directed learning
26	8	9	21	2025-03-13 18:00:00	2025-03-13 19:00:00	2	2025-03-28 00:52:18	2	Student union activity	Rejected	Conference Coordinator 1	Club Meeting
29	5	2	12	2025-04-01 08:00:00	2025-04-02 08:00:00	1	2025-04-17 23:53:00	1	For discussion	agree	Conference Coordinator 1	CP202 Discussion
30	6	9	16	2025-04-13 03:00:00	2025-04-13 06:00:00	0	2025-04-14 21:12:29	0	Self-directed learning			Self-directed learning
31	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
32	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
33	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
34	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
35	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
36	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
37	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
38	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
39	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
40	6	9	16	2025-04-22 03:00:00	2025-04-22 06:00:00	0	2025-04-21 03:00:00	0	Self-directed learning			Self-directed learning
41	1	4	13	2025-04-18 23:00:00	2025-04-19 00:00:00	0	2025-04-19 02:16:13	0	Group Discussion	(Null)	(Null)	Group Discussion
43	1	4	13	2025-04-21 22:00:00	2025-04-21 23:00:00	1	2025-04-20 20:31:54	1	Test1	Agree	Administrator	Test1
44	1	4	11	2025-04-23 22:00:00	2025-04-24 06:00:00	0	2025-04-20 21:15:01	0	Test2	(Null)	Conference Coordinator 1	Test2
46	1	4	13	2025-04-22 22:00:00	2025-04-23 01:00:00	1	2025-04-21 17:37:02	1	test	ok	Administrator	test
47	1	4	44	2025-04-24 22:00:00	2025-04-25 01:00:00	1	2025-04-23 23:01:26	1	1111	1111	Administrator	111

APPENDIX 16 (2) : Schematic diagram of some key data tables of Navicat--meeting_category

id # bigint	category_name varchar(255)	order_num varchar(255)
1	Standard Meeting Room	1
2	Multimedia Meeting Ro	2
4	Group Discussion Room	3
5	Silent Cabin	4
6	Group Discussion Area	5
7	Lecture Hall	6

APPENDIX 16 (3) : Schematic diagram of some key data tables of Navicat--meeting_room

id # bigint	name varchar(255)	area # int	category_id # int	address varchar(255)	contain # int	device text	image varchar(255)	status # tinyint
1	Thought Pavilion	50	4	SIP-North Campus-SD373	12	- Smart collaborative display /	http://47.100.1.155:8089/images/	1
2	Wisdom Chamber	20	6	SIP-North Campus-Area 1 (SE) on FB5F	10	- Mobile group tables and che	http://47.100.1.155:8089/images/	1
3	Inspiration Studio	20	6	SIP-North Campus-Area 2 (NE) on FB5F	10	- Mobile group tables and che	http://47.100.1.155:8089/images/	1
4	Enlightenment Hall	100	7	SIP-North Campus-MA105	100	- HD projector or LED display-	http://47.100.1.155:8089/images/	1
5	Knowledge Hub	100	4	SIP-South Campus-BS4114	50	- Smart collaborative display /	http://47.100.1.155:8089/images/	1
6	Discernment Study	10	5	SIP-North Campus-FB145	6	- Soundproof walls or booth d	http://47.100.1.155:8089/images/	1
7	Diligent Study	120	2	SIP-North Campus-FB251	100	- HD projector / LED video wa	http://47.100.1.155:8089/images/	1
8	Philosophy Room	100	1	SIP-South Campus-HSG26	50	- Large monitor / basic projec	http://47.100.1.155:8089/images/	1
10	Logic Lounge	10	5	SIP-North Campus-FB147	6	- Soundproof walls or booth d	http://47.100.1.155:8089/images/	1
11	Harmony Chamber	150	7	SIP-South Campus-IRG16	200	- HD projector or LED display-	http://47.100.1.155:8089/images/	1
12	Test	100	7	SIP-SA	200	Test	http://47.100.1.155:8089/images/	1

APPENDIX 16 (4) : Schematic diagram of some key data tables of Navicat--repair

id # bigint	content_text nvarchar(255)	user_id # int	create_time datetime	image nvarchar(255)	room_id # int	status nvarchar(2)
4	Projector Not Working	1	2025-02-17 08:46:10	http://47.100.1.155:8089/images/	1	1
5	Speakers No Sound	1	2025-02-17 13:18:19	http://47.100.1.155:8089/images/	2	0
6	Screen Broken	1	2025-02-18 17:00:17	http://47.100.1.155:8089/images/	6	1

APPENDIX 16 (5) : Schematic diagram of some key data tables of Navicat--sys_dept

id # bigint	dept_name nvarchar(64)	order_num # int
1	School of Science	7
2	Design School	2
3	School of Humanities and Social Sciences	3
4	International Business School Suzhou	4
5	School of Languages	5
6	School of Mathematics and Physics	6
9	School of Advanced Technology	1
10	Chinese Cultural Teaching Centre	8
11	Physical Education Centre	9
12	Information Technology Management	10

APPENDIX 16 (6) : Schematic diagram of some key data tables of Navicat--sys_menu

menu_id # int	parent_id # int	title nvarchar(36)	code nvarchar(36)	name nvarchar(36)	path nvarchar(36)	url nvarchar(128)	type nvarchar(2)	icon nvarchar(36)	parent_name nvarchar(36)	order_num # int
2	0	User Management	sysuserindex	userList	/userList	views/system/User/User	1	User	Top-level Menu	7
3	0	Role Management	sysroleindex	roleList	/roleList	views/system/Role/Role	1	Key	Top-level Menu	8
4	0	Menu Management	sysmenuindex	menuList	/menuList	views/system/Menu/Menu	1	Menu	Top-level Menu	9
8	2	Add	sysuseradd				2		User Management	21
11	2	Edit	sysuseredit				2		User Management	22
12	2	Delete	sysuserdelete				2		User Management	23
13	3	Add	sysroleadd				2		Role Management	31
14	3	Edit	sysroleedit				2		Role Management	32
15	3	Delete	sysroledelete				2		Role Management	33
16	4	Add	sysmenuadd				2		Menu Management	41
17	4	Edit	sysmenuedit				2		Menu Management	42
18	4	Delete	sysmenudelete				2		Menu Management	43
19	3	Assign Permissions	sysroleassign				2		Role Management	34
20	2	Reset Password	sysuserreset				2		User Management	24
23	2	Disable User	sysuserenabled				2		User Management	25
26	0	Department Management	sysdeptlist	dept	/dept	views/sys_dept/index.v	1	School	Top-level Menu	10

APPENDIX 16 (7) : Schematic diagram of some key data tables of Navicat--sys_role

role_id # int	role_name ρΒϵ varchar(36)	type ρΒϵ varchar(4)	remark ρΒϵ varchar(64)	create_time ⌚ datetime	update_time ⌚ datetime
13	System Administrator	(Null)	Full access to system settings and user management.	2025-03-10 08:00:00	2025-03-23 20:33:00
14	Faculty Member	(Null)	Teaching staff with meeting management permissions.	2025-02-17 08:00:00	2025-03-23 20:33:27
16	Student	(Null)	Can book meeting rooms and view notices.	2025-02-17 08:00:00	2025-03-23 20:33:34
18	Repair Staff	(Null)	Responsible for maintenance tasks.	2025-02-18 08:00:00	2025-03-23 20:33:47
19	Visitor	(Null)	Limited viewing access for external guests.	2025-03-23 20:34:32	(Null)
20	Information Technology Staff	(Null)	Responsible for IT tasks.	2025-03-27 08:00:00	2025-03-27 21:26:13

APPENDIX 16 (8) : Schematic diagram of some key data tables of Navicat--sys_user

user_id # int	dept_id # int	username ρΒϵ varchar(100)	password ρΒϵ varchar(128)	phone ρΒϵ varchar(18)	email ρΒϵ varchar(50)	sex ρΒϵ varchar(4)	is_admin ρΒϵ varchar(2)
1	3	admin	\$2a\$10\$K/oZ8.qNgU.7f	*****	*****	0	1
6	4	2255403	\$2a\$10\$WYt7yYQArKZI	123456789	Xinyuan.Zhang2202@st	1	0
8	4	2255986	\$2a\$10\$7jYUv.fxqgRVV	123456789	Wang2202@student.xjt	1	0
9	4	2251997	\$2a\$10\$vbNgzAKTv4rk	123456789	Xinrong.Shi22@student	1	0
10	4	2251903	\$2a\$10\$QcbgDLAEfwxi	123456789	Hanqiu.Zhang2202@sti	1	0
11	4	2255207	\$2a\$10\$yMyJm1NFzgx	123456789	Xiaoya.Li22@student.xji	1	0
12	4	2252173	\$2a\$10\$RAF2atKQ2DL	123456789	Nan.Lin22@student.xjtl	0	0
13	4	2147369	\$2a\$10\$A7nBAxZTm.rg	123456789	Zimo.Zheng21@studen	0	0
14	4	2251651	\$2a\$10\$MJXQciR7Oqs	18913545693	Xiangyi.Zhu22@studen	1	0
15	4	2253073	\$2a\$10\$OLeCu9EmYwq	123456789	Ziwei.Zhou22@student.	1	0

APPENDIX 17 (1) : Technical Documentation--Development Tools

Development Tools

Version Control:

- Git (Code Management)
- Coding.net (Remote Repository)

Project Management:

- Coding.net (Scrum Task Management)
- Wiki (Document Collaboration)

Database Management:

- Navicat (11.1.13) (Visual Database Management)

IDE & Editors:

- Visual Studio Code (1.98.0) (Frontend & Backend Development)

Documents & Notes:

- Word (Official Documentation)

Programming Languages

Frontend:

- HTML, CSS (Basic Page Structure and Styling)
- JavaScript (Frontend Interaction)
- TypeScript (Enhancing Code Robustness)

Backend:

- Java 17 (17.0.14)

Main Frameworks & Libraries

Frontend:

- Vue 3 (3.3.4) (Frontend Framework)
- Vite (4.4.5) (Build Tool)
- Vue Router (4.2.4) (Routing Management)
- Element Plus (2.3.12) (UI Component Library)
- Axios (1.7.2) (Frontend API Requests)
- ECharts (5.4.3) (Charts and Graphs)
- FullCalendar + Plugin Set (6.1.15) (Calendar Component)
- Vue-Tippy (6.6.0) (Tooltip Component)

Backend:

- Spring Boot (3.2.7) (Main Framework)
- Maven (3.8.1) (Dependency Management and Packaging Tool)
- Druid (1.2.1) (Database Connection Pool)
- MyBatis-Plus + MyBatis-Plus-Join (3.5.7 + 1.4.7.2) (ORM Framework)
- Sa-Token (Authentication and Authorization) (1.38.0)
- Fastjson (1.2.68) (JSON Processing Tool)
- Spring Security Crypto, JWT (Java-JWT) (6.1.4 + 3.10.3) (Security Components)

Database:

- MySQL (5.7.27) (Relational Database)

Authentication:

- JWT (JSON Web Token)

Testing Frameworks:

- JUnit (Java Unit Testing)
- Mockito (Unit Testing Mock)

Required Knowledge

Software Engineering & Development Process

- Scrum Agile Development
- Sprint Planning, Iterative Development, Daily Standups
- PBI (Product Backlog Item) Management
- Task Assignment (Epic / Story / Task)

Database Design

- Database Table Design (ERD Diagrams)
- Relational Database
- SQL Query Optimization (Indexes, Transactions)

Frontend Technologies

- Vue 3 Component-Based Development
- Vue Router Page Navigation Control
- Axios Network Request Interception
- ECharts Data Visualization
- Element Plus Form Validation and Dialog Boxes

Backend Technologies

- Spring Boot Controller-Service-Mapper Layered Architecture
- MyBatis-Plus Dynamic Conditional Query Encapsulation
- Sa-Token Login Authentication and Role-Based Access Control
- Email Registration & Verification Code Sending Logic
- Image Uploading and Static Resource Management (Local Path Mapping)

Deployment and Delivery

- Development Environment:
- Local Running + Vite Hot Reload + Maven Build
- Server Deployment:
- Cloud Server (Alibaba Cloud)
- Nginx Reverse Proxy
- Node.js Environment Version: 18.20.8
- Frontend Deployment:
- Deployment to the dist Directory on the Cloud Server
- Backend Deployment:
- Upload and Run Packaged .jar File (Spring Boot Standalone Application)

Suggested Workflow

Project Initialization

- Initialize frontend and backend using Vue + Vite + Spring Boot
- Set up Git project structure: main / develop / student ID branches
- Configure a unified API response format and exception handling mechanism

Requirements Analysis & Task Assignment

- Record requirements using Wiki
- Assign tasks via Coding.net
- Create ER diagrams based on user stories
- Plan functional modules, interface prototypes, and permission/role systems

Development Stage

- Frontend: Interface development, database interactions, validation logic encapsulation
- Backend: UI development, API calls, state management synchronization

Testing and Integration

- Console Debugging, User Behavior Tracking (using Sentry)

Deployment

- Package backend into a .jar file
- Build frontend with Vite and upload to server